AOS-CX 10.16.xxxx ACLs and Classifier Policies Guide

(8400 Switch Series)

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Notices

Contents

Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
`example-text`	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (**[ ]**).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ *`<example-text>`* ■ `<example-text>` ■ *`example-text`* ■ *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (**< >**). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
`|`	Vertical bar. A logical **OR** that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
`{ }`	Braces. Indicates that at least one of the enclosed items is required.

Convention	Usage
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or . . .	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where *<VLAN-ID>* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: ***member/slot/port***.

On the HPE Aruba Networking 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/5 and 1/6.
  - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
  - *member*: 1.
  - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
  - *member*: 1.
  - *tray*: 1 to 4.
  - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
  - *member*: 1.
  - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Access Control Lists

Access Control Lists (ACLs) let a network administrator permit or deny passage of traffic based on network addresses, protocols, service ports, and other packet attributes. ACLs are composed of one or more Access Control Entries (called ACEs). Each ACE defines a filter criteria and an action, either **permit** or **deny**. If the traffic matches the filter criteria, the specified action is taken. The **permit** action permits the traffic to continue through the switch. The **deny** action causes the traffic to be discarded (dropped). ACEs can also log or count matching traffic.

Three ACL types are supported; IPv4, IPv6, and MAC. Each ACL type is focused on relevant frame or packet characteristics.

ACLs must be applied (using an **apply access-list** command) to take effect. ACLs can be applied to interfaces (including LAGs), VLANs, or the Control Plane.

Access Control Entries (ACEs) are listed according to priority by sequence number and processed in lowest to highest sequence number order. Each ACE attempts to match on one or more attributes of the particular traffic type. Attempted ACE matching ceases upon the first successful match. For a match to be considered successful, a packet must match all the criteria, qualifiers, and attributes of a particular ACE. Higher-numbered ACEs are only processed if no lower-numbered ACE matches. If the traffic matches no ACE in the entire ACL, the default action **deny** is taken, causing the traffic to be discarded (dropped).

When defining an ACE, if the sequence number is omitted, the ACE is auto-assigned a new sequence number that is 10 greater than the existing highest ACE sequence number. The first auto-assigned sequence number is 10. If you choose to include the ACE sequence numbers, you can use any number you like, however it is suggested that you follow the practice of entering them as 10, 20, 30, and so on. Regardless of the order in which ACEs are entered, they are stored in low-to-high sequence number order. If you enter three ACEs numbered 10, 30, 20, when creating an ACL, the ACEs are stored in the ACL as 10, 20, 30.

This simple ACL definition permits traffic passage for a particular address range and otherwise counts all nonmatching (dropped) traffic:

```
switch(config)# access-list ip network-A-udp-only
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 deny any any any count
switch(config-acl-ip)# exit
```

The main traffic characteristics that ACEs can filter on are as follows (see the full list in the ACE parameters list of the ACL commands):

- Protocol such as: ICMP, TCP, UDP
- Source and/or destination addresses (IPv4, IPv6, or MAC)
- Source and/or destination TCP/UDP ports (if applicable to the specified protocol)

A few real-world uses of ACLs are as follows:

- Restrict traffic arriving on a routed port, destined to a particular address or subnet by applying an ACL that matches on a destination IP address or an IP address and a mask.
- Prevent an entire subnet from routing through a port by applying an ACL that matches on IP source address and a mask.
- Prevent certain protocols from using a particular multicast MAC address (advertising through a port) by applying an ACL that matches on the destination MAC address.
- Prevent any IP host from accessing a particular IP port/application on a specific server by applying an ACL that matches on IP addresses and Layer 4 port.

See also ACL and Policy hardware resource considerations.

ACL usage tips

When using the `access-list ip` or `access-list ipv6` commands, if you enter an existing `ACL-NAME`, the existing ACL is modified as follows:

- Any ACE entered with a new sequence-number creates an additional ACE.
- Any ACE entered with an existing sequence-number replaces the existing ACE.

If you modify an ACL that has already been applied, it is possible that packets, blocked by the previous ACL, will briefly pass through the switch during the ACL reconfiguration.

In a highly secure environment, it is safest to first bring down interfaces and VLANs to which an ACL has been applied before modifying the ACL. Then bring the targets of ACL application back up after completing the ACL modification. Respecting this recommendation ensures that an ACL is never partially programmed while traffic is passing through the switch.

About applying ACLs to interfaces or LAGs

You can apply an ACL to an interface or LAG to affect or control the traffic arriving on that interface or LAG (inbound) or leaving the interface or LAG (outbound), or both. A given interface or LAG supports the application of a single ACL per type, per direction. ACLs can be applied to interfaces or LAGs as follows:

- One MAC ACL inbound
- One IPv4 ACL inbound
- One IPv4 ACL outbound
- One IPv6 ACL inbound

Different ACLs of the same type can be used in opposite directions for IPv4. If you apply an ACL of a particular type, in a direction that is already in use, the switch replaces the current ACL with the new ACL.

About applying ACLs to VLANs

ACLs can be applied to VLANs only in the inbound (ingress) direction.

Sequence numbering

If no sequence number is specified, the software appends new ACEs to the end of the ACL with a sequence number equal to the highest ACE currently in the list plus 10.

The sequence numbers may be resequenced using the `access-list resequence` command.

Deny ACLs

If multiple ACLs of different types are applied in the same direction, a deny ACE, whether explicit or implicit, in one ACL overrides a permit ACL in another. A deny ACE is an ACE within an ACL that uses the `deny` action keyword.

Denied ping requests

A ping request is denied when an ACL is applied on ingress or egress unless the request is explicitly permitted.

```
switch# ping 100.1.2.10
PING 100.1.2.10 (100.1.2.10) 100(128) bytes of data.
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
```

ACL failure behaviors

- In the event of a failed ACL application to a VLAN or VLAN interface during a hotswap or switch reboot, all of the ports will shut down. The switch, line card or stack member must be restarted to recover from the failure. Modifying the VLAN, VLAN interface, or ACL configuration will not cause the switch, line card or stack member to be restarted.

ACLs and VXLAN Traffic

ACLs are not supported for VXLAN traffic on 8400 switch series.

About address and port object groups

Object groups are useful for defining groups of IP addresses and Layer 4 ports for use exclusively in the two ACL-defining commands `access-list ip` and `access-list ipv6`.

Often, common groups of addresses and ports or port ranges are used repeatedly in many ACL definitions. Without address and port object groups, the same addresses and ports must be repeated in each ACL definition that uses them.

With address and port object groups, the IP addresses and ports can be defined once, using any of these commands:

- `object-group ip address`
- `object-group ipv6 address`
- `object-group port`

Once an object group is defined, the group is available for inclusion by name as the *<ADDRESS-GROUP>* and *<PORT-GROUP>* parameters in the `access-list ip` and `access-list ipv6` ACL-definition commands.

Object groups simplify the ACL definition process and help ensure consistent address and port specification across many ACLs.

Keep in mind that it is possible to consume many hardware resource entries when using the object group commands. For example, in a typical situation, an ACE that uses object groups with 3 source addresses, 3 source L4 ports, 3 destination addresses, and 3 destination L4 ports, a total of 81 hardware entries are consumed (3 * 3 * 3 * 3 = 81).

ACL and ACE-related tasks

Common ACL and ACE-related tasks are as follows:

Task	Command name	Example

Creating an IPv4 ACL	`access-list ip`	`access-list ip MY_IP_ACL`
		`  10 permit udp any 172.16.1.0/24`
		`  20 permit tcp 172.16.2.0/16 gt 1023 any`
		`  30 deny any any any count`

Creating an IPv6 ACL	`access-list ipv6`	`access-list ipv6 MY_IPV6_ACL`
		`  10 permit udp any 2001::1/64`
		`  20 permit tcp 2001:2011::1/64 any`
		`  30 deny any any any count`

Creating a MAC ACL	`access-list mac`	`access-list mac MY_MAC_ACL`
		`  10 permit any any appletalk vlan 40`
		`  20 deny any any any count`

Applying an IPv6 ACL to an interface	`apply access-list` (to interface or LAG)	`interface 1/1/1` `  apply access-list ipv6 MY_IPV6_ACL in`
Applying an IPv4 ACL to a LAG	`apply access-list` (to interface or LAG)	`interface lag 100` `  apply access-list ip MY_IP_ACL in`
Applying an IPv4 ACL to a VLAN	`apply access-list` (to VLAN)	`vlan 10` `  apply access-list ip MY_IP_ACL in`
Applying a MAC ACL to a VLAN	`apply access-list` (to VLAN)	`vlan 40` `  apply access-list mac MY_MAC_ACL in`
Applying an IPv4 ACL to the Control Plane (OOBM)	`apply access-list control-plane`	`apply access-list ip MY_IP_ACL control-plane vrf mgmt`
Removing application of an ACL from an interface	`apply access-list` (to interface or LAG)	`interface 1/1/1` `  no apply access-list ipv6 MY_IPV6_ACL in`
Removing application of an ACL from a VLAN	`apply access-list` (to VLAN)	`vlan 40` `  no apply access-list mac MY_MAC_ACL in`
Removing application of an ACL from the Control Plane (OOBM)	`apply access-list control-plane`	`no apply access-list ip MY_IP_ACL control-plane vrf mgmt`
Showing all ACLs	`show access-list`	`show access-list`
Showing all IPv6 ACLs	`show access-list`	`show access-list ipv6`
Showing all ACLs applied to interface `1/1/1`	`show access-list`	`show access-list interface 1/1/1`
Showing all ACLs applied to VLAN `10`	`show access-list`	`show access-list vlan 10`
Showing all ACLs applied to the Control Plane	`show access-list control-plane`	`show access-list control-plane`

Task	Command name	Example
Showing a particular ACL	`show access-list`	`show access-list ip MY_ACL`
Showing an ACL as commands	`show access-list`	`show access-list ip MY_ACL commands`
Showing ACL hit counts for an ACL applied to an interface	`show access-list hitcounts`	`show access-list hitcounts ip MY_ACL interface 1/1/1`
Showing ACL hit counts for an ACL applied to a VLAN	`show access-list hitcounts`	`show access-list hitcounts ip MY_ACL vlan 10`
Showing ACL hit counts for an ACL applied to the Control Plane	`show access-list hitcounts control-plane`	`show access-list hitcounts ip MY_ACL control-plane vrf mgmt`
Clearing ACL hit counts	`clear access-list hitcounts`	`clear access-list hitcounts ip MY_ACL vlan 10`
Clearing ACL hit counts for Control Plane	`clear access-list hitcounts control-plane`	`clear access-list hitcounts control-plane vrf mgmt`
Copying an ACL	`access-list copy`	`access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2`
Resequencing the ACEs of an ACL	`access-list resequence`	`access-list ip MY_IP_ACL resequence 1 1`
Resetting an ACL	`access-list reset`	`access-list ip MY_IP_ACL reset`
Setting the ACL log timer frequency	`access-list log-timer`	`access-list log-timer 30`

Object group-related tasks

Object groups are useful for defining groups of addresses and ports for use exclusively in the two ACL-defining commands `access-list ip` and `access-list ipv6`.

Common object group-related tasks are as follows:

Task	Command name	Example
Creating an IPv4 address object group	`object-group ip address`	`object-group ip address my_ipv4_addr_group`
Creating an IPv6 address object group	`object-group ipv6 address`	`object-group ipv6 address my_ipv6_addr_group`
Creating a port	`object-group`	`object-group port my_port_group`

Task	Command name	Example
object group	port	
Showing an IPv4 address object group	show object-group	show object-group ip address my_ipv4_addr_group
Showing all IPv6 address object groups	show object-group	show object-group ipv6 address
Showing a port object group	show object-group	show object-group port my_port_group
Showing all port object groups as commands	show object-group	show object-group port commands
Resequencing an IPv4 address object group	object-group ip address	object-group ip address my_ipv4_addr_group resequence 100 10
Resequencing a port object group	object-group port	object-group port my_port_group resequence 200 5
Resetting an IPv6 address object group	object-group ipv6 address	object-group ipv6 address my_ipv6_addr_group reset
Resetting a port object group	object-group port	object-group port my_port_group reset

Active ACL configuration versus user-specified configuration

The **show access-list** command shows the active configuration of the switch. The active configuration is the ACLs that have been configured and accepted by the system. The active configurations are the interfaces on which the ACLs have successfully been programmed in the hardware.

The output of the `show access-list` command with the **configuration** parameter shows the ACLs that have been configured. The output of this command may not be the same as what was programmed in the hardware or what is active on the switch. The situation might occur because of one or more of the following:

- Unsupported command parameters might have been configured.
- Unsupported applications might have been specified.
- Applying an ACL might have been unsuccessful due to lack of hardware resources.

To determine if a discrepancy exists between what was configured and what is active, run the `show access-list` command with the `configuration` parameter.

If the active ACLs and configured ACLs are not the same, the switch shows a warning message in the output of the show command:

```
! access-list ip MY_IP_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
```

If the configured ACL is processing, the switch shows an in-progress warning.

```
! access-list ip MY_IP_ACL user configuration currently being processed
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
```

If the switch shows a warning message or in-progress message, additional changes can be made until the error message is no longer shown in the show command, or you can run the `access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset` command. The `access-list reset` command changes the user-specified configuration to match the active configuration. For details, see access-list reset.

The `show running-config` command also shows a warning about ACLs that are in progress or failed.

Examples

Applying an ACL with TCP acknowledgments (ACKs) on ingress, which is unsupported by the hardware:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Showing the user-specified configuration:

```
switch(config)# do show access-list ip TEST_ACL
        10 permit tcp 172.16.2.0/16 any ack
    interface 1/1/1
    ! access-list ip TEST_ACL user configuration does not match active
configuration.
    ! run 'show access-list [commands]' to display active access-list
configuration.
        apply access-list ip TEST_ACL in

switch(config)# do show access-list commands
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/16 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list all reset' to reset all access-lists to match active
configuration.

switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/16 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list all reset' to reset all access-lists to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
```

```
            Source IP Address                   Source L4 Port(s)
            Destination IP Address              Destination L4 Port(s)
            Additional Parameters
-------------------------------------------------------------------------------
IPv4        TEST_ACL
         10 permit                              tcp
            172.16.2.0/16
            any
            ack
```

Resetting the user-specified configuration to match the active configuration:

```
switch(config)# access-list all reset
```

Showing the updated user-specified configuration:

```
switch(config)# do show access-list commands configuration
   access-list ip TEST_ACL
       10 permit tcp 172.16.2.0/16 any ack
```

ACL commands

ACL application

ACLs can be applied as follows:

ACL type Direction	IPv4+6 In	IPv4 Out	MAC In
L2 interface (port)	Yes		Yes
L2 LAG	Yes		Yes
L3 interface (port)	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes
VLAN	Yes		Yes
Interface VLAN	Yes (routed)		
Management interface	Yes		
Control Plane (per VRF)	Yes		

Egress ACLs can only be applied to L3 (route-only) interfaces. Applying an egress ACL to an L2 interface will result in an error. Only ingress ACLs (IPv4, IPv6, and MAC) can be applied to VLANs. Applying an egress ACL to VLAN will result in an error.

The following match criteria are not supported. If any of these match criteria are attempted to be configured, an error message will be displayed and the action will not be completed.

```
TCP flags CWR and ECE
TCP flags and TTL (hop limit) on IPv6 ACLs
TCP flags and fragment on outbound ACLs
Fragment on IPv6 VLAN ACLs
VLAN ID on VLAN ACLs
```

To apply IPv4 and/or IPv6 ACLs to the management interface, apply them to the Control Plane on the management VRF.

access-list copy

`access-list {ip|ipv6|mac} <ACL-NAME> copy <DESTINATION-ACL>`

Description

Copies an IPv4, IPv6, or MAC ACL to a new destination ACL or overwrites an existing ACL.

Parameter	Description
`{ip|ipv6|mac}`	Specifies the type of ACL.
<ACL-NAME>	Specifies the name of the ACL to be copied.
<DESTINATION-ACL>	Specifies the name of the destination ACL.

Examples

Copying *MY_IP_ACL* to *MY_IP_ACL2*:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
         1 permit                          udp
           any
           172.16.1.0/255.255.255.0
         2 permit                          tcp
           172.16.2.0/255.255.0.0           >  1023
           any
         3 permit                          tcp
           172.26.1.0/255.255.255.0
```

```
          any
          dscp: AF11
          ack
          syn
        4 deny                          any
          any
          any
          Hit-counts: enabled
-------------------------------------------------------------------------------
IPv4      MY_IP_ACL2
        1 permit                        udp
          any
          172.16.1.0/255.255.255.0
        2 permit                        tcp
          172.16.2.0/255.255.0.0         >  1023
          any
        3 permit                        tcp
          172.26.1.0/255.255.255.0
          any
          dscp: AF11
          ack
          syn
        4 deny                          any
          any
          any
          Hit-counts: enabled
```

Copying MY_IPV6_ACL to MY_IPV6_ACL2:

```
switch(config)# access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
  Sequence Comment
          Action                        L3 Protocol
          Source IP Address             Source L4 Port(s)
          Destination IP Address        Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv6      MY_IPV6_ACL
        1 permit                        udp
          any
          2001::1/64
        2 Permit all TCP ephemeral ports
          permit                        tcp
          2001:2001::2:1                 >  1023
          any
        3 permit                        tcp
          2001:2011::1/64
          any
        4 deny                          any
          any
          any
          Hit-counts: enabled
-------------------------------------------------------------------------------
IPv6      MY_IPV6_ACL2
        1 permit                        udp
          any
          2001::1/64
        2 Permit all TCP ephemeral ports
```

```
          permit                          tcp
          2001:2001::2:1                   >  1023
          any
        3 permit                          tcp
          2001:2011::1/64
          any
        4 deny                            any
          any
          any
          Hit-counts: enabled
```

Copying MY_MAC_ACL to MY_MAC_ACL2:

```
switch(config)# access-list mac MY_MAC_ACL copy MY_MAC_ACL2
switch(config-acl-mac)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        1 permit                          ipv6
          1122.3344.5566/ffff.ffff.0000
          any
        2 permit                          any
          aaaa.bbbb.cccc
          1111.2222.3333
          QoS Priority Code Point: 4
        3 Permit all vlan-1 tagged Appletalk traffic
          permit                          appletalk
          any
          any
          VLAN: 1
        4 deny                            any
          any
          any
          Hit-counts: enabled
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL2
        1 permit                          ipv6
          1122.3344.5566/ffff.ffff.0000
          any
        2 permit                          any
          aaaa.bbbb.cccc
          1111.2222.3333
          QoS Priority Code Point: 4
        3 Permit all vlan-1 tagged Appletalk traffic
          permit                          appletalk
          any
          any
          VLAN: 1
        4 deny                            any
          any
          any
          Hit-counts: enabled
Type       Name
  Sequence Comment
```

```
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
         1 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
         2 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
         3 Permit all vlan-1 tagged Appletalk traffic
           permit                          appletalk
           any
           any
           VLAN: 1
         4 deny                            any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL2
         1 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
         2 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
         3 Permit all vlan-1 tagged Appletalk traffic
           permit                          appletalk
           any
           any
           VLAN: 1
         4 deny                            any
           any
           any
           Hit-counts: enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

access-list ip

Syntax to create an IPv4 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ip <ACL-NAME>
no access-list ip <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols **ah**, **gre**, **esp**, **igmp**, **ospf**, **pim** (**ip** is available as an alias for **any**):

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|<ADDRESS-GROUP>}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>]
[count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols `sctp`, `tcp`, `udp`:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>]  [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>]
[count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol `icmp`:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmp}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|<ADDRESS-GROUP>}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>]
[count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates an IPv4 Access Control List (ACL) comprised of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Parameter	Description
<ACL-NAME>	Specifies the name of this ACL.
<SEQUENCE-NUMBER>	Specifies a sequence number for the ACE. Range: 1 to 4294967295.

Parameter	Description
{permit|deny}	Specifies whether to permit or deny traffic matching this ACE.
<IP-PROTOCOL-NUM>	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
{any|*<SRC-IP-ADDRESS>*[/{*<PREFIX-LENGTH>*|*<SUBNET-MASK>*}]|*<ADDRESS-GROUP>*}	Specifies the source IPv4 address. ▪ **any** - specifies any source IPv4 address. ▪ ***<SRC-IP-ADDRESS>*** - specifies the source IPv4 host address. ◦ ***<PREFIX-LENGTH>*** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ ***<SUBNET-MASK>*** - specifies the address bits to mask (dotted decimal notation). ▪ ***<ADDRESS-GROUP>*** - specifies an IPv4 address group defined with **object-group ip address**.
{any|*<DST-IP-ADDRESS>*[/{*<PREFIX-LENGTH>*|*<SUBNET-MASK>*}]|*<ADDRESS-GROUP>*}	Specifies the destination IPv4 address. ▪ **any** - specifies any destination IPv4 address. ▪ ***<DST-IP-ADDRESS>*** - specifies the destination IPv4 host address. ◦ ***<PREFIX-LENGTH>*** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ ***<SUBNET-MASK>*** - specifies the address bits to mask (dotted decimal notation). ▪ ***<ADDRESS-GROUP>*** - specifies an IPv4 address group that you defined earlier with object-group ip address.
[{eq|gt|lt} *<PORT>*|range *<MIN-PORT>* *<MAX-PORT>*|group *<PORT-GROUP>*]	Specifies the port, port range, or port group. Port numbers are in the range of 0 to 65535. ▪ **eq *<PORT>*** - specifies the Layer 4 port. ▪ **gt *<PORT>*** - specifies any Layer 4 port greater than the indicated port. ▪ **lt *<PORT>*** - specifies any Layer 4 port less than the indicated port. ▪ **range *<MIN-PORT>* *<MAX-PORT>*** - specifies the Layer 4 port range. ▪ **group *<PORT-GROUP>*** - specifies the Layer 4 port group that you defined earlier with object-group port. **NOTE:** Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.
urg	Specifies matching on the TCP Flag: **Urgent**. (Applies

Parameter	Description
	only to the "in" (ingress) direction.)
`ack`	Specifies matching on the TCP Flag: **Acknowledgment**. (Applies only to the "in" (ingress) direction.)
`psh`	Specifies matching on the TCP Flag: **Push buffered data to receiving application**. (Applies only to the "in" (ingress) direction.)
`rst`	Specifies matching on the TCP Flag: **Reset the connection**. (Applies only to the "in" (ingress) direction.)
`syn`	Specifies matching on the TCP Flag: **Synchronize sequence numbers**. (Applies only to the "in" (ingress) direction.)
`fin`	Specifies matching on the TCP Flag: **Finish connection**. (Applies only to the "in" (ingress) direction.)
`established`	Specifies matching on the TCP Flag: Established connection. (Applies only to the "in" (ingress) direction.)
`[icmp-type {echo|echo-reply|` `<ICMP-TYPE-VALUE>}]`	Specifies the ICMP type. ■ **echo** - specifies an ICMP echo request packet. ■ **echo-reply** - specifies an ICMP echo reply packet. ■ ***<ICMP-TYPE-VALUE>*** - specifies an ICMP type value. Range: 0 to 255.
`[icmp-code <ICMP-CODE-VALUE>]`	Specifies the ICMP code value. Range: 0 to 255.
`dscp DSCP-SPECIFIER>`	Specifies the Differentiated Services Code Point (DSCP), either a numeric ***<DSCP-VALUE>*** (0 to 63) or one of these keywords: ■ **AF11** - DSCP 10 (Assured Forwarding Class 1, low drop probability) ■ **AF12** - DSCP 12 (Assured Forwarding Class 1, medium drop probability) ■ **AF13** - DSCP 14 (Assured Forwarding Class 1, high drop probability) ■ **AF21** - DSCP 18 (Assured Forwarding Class 2, low drop probability) ■ **AF22** - DSCP 20 (Assured Forwarding Class 2, medium drop probability) ■ **AF23** - DSCP 22 (Assured Forwarding Class 2, high drop probability) ■ **AF31** - DSCP 26 (Assured Forwarding Class 3, low drop probability) ■ **AF32** - DSCP 28 (Assured Forwarding Class 3, medium drop probability)

Parameter	Description
	■ **AF33** - DSCP 30 (Assured Forwarding Class 3, high drop probability) ■ **AF41** - DSCP 34 (Assured Forwarding Class 4, low drop probability) ■ **AF42** - DSCP 36 (Assured Forwarding Class 4, medium drop probability) ■ **AF43** - DSCP 38 (Assured Forwarding Class 4, high drop probability) ■ **CS0** - DSCP 0 (Class Selector 0: Default) ■ **CS1** - DSCP 8 (Class Selector 1: Scavenger) ■ **CS2** - DSCP 16 (Class Selector 2: OAM) ■ **CS3** - DSCP 24 (Class Selector 3: Signaling) ■ **CS4** - DSCP 32 (Class Selector 4: Real time) ■ **CS5** - DSCP 40 (Class Selector 5: Broadcast video) ■ **CS6** - DSCP 48 (Class Selector 6: Network control) ■ **CS7** - DSCP 56 (Class Selector 7) ■ **EF** - DSCP 46 (Expedited Forwarding)
`ecn` *`<ECN-VALUE>`*	Specifies an Explicit Congestion Notification value. Range: 0 to 3.
`ip-precedence` *`<IP-PRECEDENCE-VALUE>`*	Specifies an IP precedence value. Range: 0 to 7.
`tos` *`<TOS-VALUE>`*	Specifies the Type of Service value. Range: 0 to 31.
`fragment`	Specifies a fragment packet.
`vlan` *`<VLAN-ID>`*	Specifies VLAN tag to match on. 802.1Q VLAN ID. **NOTE:** This parameter cannot be used in any ACL that will be applied to a VLAN.
`ttl <TTL-VALUE>`	Specifies a time-to-live (hop limit) value. Range: 0 to 255.
`count`	Keeps the hit counts of the number of packets matching this ACE.
`log`	Keeps a log of the number of packets matching this ACE. Works with `deny` actions but not with `permit` actions. Works with ACLs applied on ingress, egress, or Control Plane.
`[`*`<SEQUENCE-NUMBER>`*`] comment` *`<TEXT-STRING>`*	Adds a comment to an ACE. The **no** form removes only the comment from the ACE.

Usage

- If the ***<IP-PROTOCOL-NUM>*** parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with **log** option is logged. Until the log-timer wait-period is over, any

packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log for all the ACLs that were matched, regardless of type.

Examples

Creating an IPv4 ACL with four entries:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 permit tcp 172.16.2.0/16 gt 1023 any
switch(config-acl-ip)# 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit

switch(config)# show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10 permit                          udp
           any
           172.16.1.0/255.255.255.0
        20 permit                          tcp
           172.16.2.0/255.255.0.0           >  1023
           any
        30 permit                          tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Adding a comment to an existing IPv4 ACE:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ip)# exit

switch(config)# show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10 permit                          udp
           any
           172.16.1.0/255.255.255.0
        20 Permit all TCP ephemeral ports
           permit                          tcp
```

```
          172.16.2.0/255.255.0.0          >  1023
          any
       30 permit                         tcp
          172.26.1.0/255.255.255.0
          any
          dscp: AF11
          ack
          syn
       40 deny                           any
          any
          any
          Hit-counts: enabled
```

Removing a comment from an existing IPv4 ACE:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 20 comment
switch(config-acl-ip)# exit

switch(config)# show access-list
Type        Name
  Sequence Comment
          Action                          L3 Protocol
          Source IP Address               Source L4 Port(s)
          Destination IP Address          Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv4        MY_IP_ACL
       10 permit                          udp
          any
          172.16.1.0/255.255.255.0
       20 permit                          tcp
          172.16.2.0/255.255.0.0           >  1023
          any
       30 permit                          tcp
          172.26.1.0/255.255.255.0
          any
          dscp: AF11
          ack
          syn
       40 deny                            any
          any
          any
          Hit-counts: enabled
```

Adding an ACE (insert line 25) to an existing IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.16.2.0/16 any
switch(config-acl-ip)# exit

switch(config)# show access-list
Type        Name
  Sequence Comment
          Action                          L3 Protocol
          Source IP Address               Source L4 Port(s)
          Destination IP Address          Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
```

```
IPv4       MY_IP_ACL
        10 permit                        udp
           any
           172.16.1.0/255.255.255.0
        20 permit                        tcp
           172.16.2.0/255.255.0.0         >  1023
           any
        25 permit                        icmp
           172.16.2.0/255.255.0.0 any
        30 permit                        tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40 deny                          any
           any
           any
           Hit-counts: enabled
```

Replacing an ACE in an existing IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.17.1.0/16 any
switch(config-acl-ip)# exit

switch(config)# show access-list
Type       Name
  Sequence Comment
           Action                        L3 Protocol
           Source IP Address             Source L4 Port(s)
           Destination IP Address        Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10 permit                        udp
           any
           172.16.1.0/255.255.255.0
        20 permit                        tcp
           172.16.2.0/255.255.0.0         >  1023
           any
        25 permit                        icmp
           172.17.1.0/255.255.0.0
        30 permit                        tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40 deny                          any
           any
           any
           Hit-counts: enabled
Type       Name
  Sequence Comment
           Action                        L3 Protocol
           Source IP Address             Source L4 Port(s)
           Destination IP Address        Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
```

```
        10 permit                         udp
           any
           172.16.1.0/255.255.255.0
        20 permit                         tcp
           172.16.2.0/255.255.0.0          >  1023
           any
        25 permit                         icmp
           172.17.1.0/255.255.0.0
        30 permit                         tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40 deny                           any
           any
           any
           Hit-counts: enabled
```

Removing an ACE from an IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 25
switch(config-acl-ip)# exit

switch(config)# show access-list
Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10 permit                         udp
           any
           172.16.1.0/255.255.255.0
        20 permit                         tcp
           172.16.2.0/255.255.0.0          >  1023
           any
        30 permit                         tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40 deny                           any
           any
           any
           Hit-counts: enabled
Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10 permit                         udp
           any
           172.16.1.0/255.255.255.0
```

```
        20 permit                          tcp
           172.16.2.0/255.255.0.0           >  1023
           any
        30 permit                          tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Copy an IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config)# show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10
           permit                          udp
           any
           172.16.1.0/255.255.255.0
        20
           permit                          tcp
           172.16.2.0/255.255.0.0           >  1023
           any
        30
           permit                          tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
        40
           deny                            any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL2
        10
           permit                          udp
           any
           172.16.1.0/255.255.255.0
        20
           permit                          tcp
           172.16.2.0/255.255.0.0           >  1023
           any
        30
           permit                          tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
```

```
          ack
          syn
       40
          deny                            any
          any
          any
          Hit-counts: enabled switch(config)# access-list ip MY_IP_ACL copy MY_
IP_ACL2

switch(config)# show access-list
Type        Name
  Sequence Comment
          Action                          L3 Protocol
          Source IP Address               Source L4 Port(s)
          Destination IP Address          Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv4        MY_IP_ACL
       10
          permit                          udp
          any
          172.16.1.0/255.255.255.0
       20
          permit                          tcp
          172.16.2.0/255.255.0.0           >  1023
          any
       30
          permit                          tcp
          172.26.1.0/255.255.255.0
          any
          dscp: AF11
          ack
          syn
       40
          deny                            any
          any
          any
          Hit-counts: enabled
-------------------------------------------------------------------------------
IPv4        MY_IP_ACL2
       10
          permit                          udp
          any
          172.16.1.0/255.255.255.0
       20
          permit                          tcp
          172.16.2.0/255.255.0.0           >  1023
          any
       30
          permit                          tcp
          172.26.1.0/255.255.255.0
          any
          dscp: AF11
          ack
          syn
       40
          deny                            any
          any
          any
          Hit-counts: enabled
```

Removing an IPv4 ACL:

```
switch(config)# no access-list ip MY_IP_ACL

switch(config)# show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL2
         1 permit                          udp
           any
           172.16.1.0/255.255.255.0
         2 permit                          tcp
           172.16.2.0/255.255.0.0           >  1023
           any
         3 permit                          tcp
           172.26.1.0/255.255.255.0
           any
           dscp: AF11
           ack
           syn
         4 deny                            any
           any
           any
           Hit-counts: enabled
```

Command History

Release	Modification
10.12	Allow ACLs applied to the Control Plane to be logged.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **access-list ip *<ACL-NAME>*** command takes you into the named ACL context where you enter the ACEs.	Administrators or local user group members with execution rights for this command.

access-list ipv6

Syntax to create an IPv6 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ipv6 <ACL-NAME>
no access-list ipv6 <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols **ah**, **gre**, **esp**, **ospf**, **pim** (**ipv6** is available as an alias for **any**):

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ipv6|ah|gre|esp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
```

```
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols `sctp`, `tcp`, `udp`:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>}]|<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>|group <PORT-GROUP>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol `icmpv6`:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>][ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates an IPv6 Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Parameter	Description
<ACL-NAME>	Specifies the name of this ACL.
<SEQUENCE-NUMBER>	Specifies a sequence number for the ACE. Range: 1 to 4294967295.
`{permit|deny}`	Specifies whether to permit or deny traffic matching this ACE.
<IP-PROTOCOL-NUM>	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
`{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]|<ADDRESS-GROUP>}`	Specifies the source IPv6 address. ■ **any** - specifies any source IPv6 address. ■ ***<SRC-IP-ADDRESS>*** - specifies the source IPv6 host

Parameter	Description
	address. ◦ ***<PREFIX-LENGTH>*** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128. ▪ ***<ADDRESS-GROUP>*** - specifies an IPv6 address group that you defined earlier with **object-group ipv6 address**.
{any|*<DST-IP-ADDRESS>*[/*<PREFIX-LENGTH>*]|*<ADDRESS-GROUP>*}	Specifies the destination IPv6 address. ▪ **any** - specifies any destination IPv6 address. ▪ ***<DST-IP-ADDRESS>*** - specifies the destination IPv6 host address. ◦ ***<PREFIX-LENGTH>*** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128. ▪ ***<ADDRESS-GROUP>*** - specifies an IPv6 address group that you defined earlier with object-group ipv6 address.
[{eq|gt|lt} *<PORT>*|range *<MIN-PORT><MAX-PORT>*|group *<PORT-GROUP>*]	Specifies the port, port range, or port group. Port numbers are in the range of 0 to 65535. ▪ **eq *<PORT>*** - specifies the Layer 4 port. ▪ **gt *<PORT>*** - specifies any Layer 4 port greater than the indicated port. ▪ **lt *<PORT>*** - specifies any Layer 4 port less than the indicated port. ▪ **range *<MIN-PORT>* *<MAX-PORT>*** - specifies the Layer 4 port range. ▪ **group *<PORT-GROUP>*** - specifies the Layer 4 port group that you defined earlier with object-group port. **NOTE:** Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.
urg, ack, psh, rst, syn, fin, established	These TCP flag-matching parameters are not supported.
[icmp-type {echo|echo-reply|*<ICMP-TYPE-VALUE>*}]	Specifies the ICMP type. ▪ **echo** - specifies an ICMP echo request packet. ▪ **echo-reply** - specifies an ICMP echo reply packet. ▪ ***<ICMP-TYPE-VALUE>*** - specifies an ICMP type value. Range: 0 to 255.
[icmp-code <ICMP-CODE-VALUE>]	Specifies the ICMP code value. Range: 0 to 255.
dscp *DSCP-SPECIFIER>*	Specifies the Differentiated Services Code Point (DSCP), either a numeric ***<DSCP-VALUE>*** (0 to 63) or one of these keywords:

Parameter	Description
	■ **AF11** - DSCP 10 (Assured Forwarding Class 1, low drop probability) ■ **AF12** - DSCP 12 (Assured Forwarding Class 1, medium drop probability) ■ **AF13** - DSCP 14 (Assured Forwarding Class 1, high drop probability) ■ **AF21** - DSCP 18 (Assured Forwarding Class 2, low drop probability) ■ **AF22** - DSCP 20 (Assured Forwarding Class 2, medium drop probability) ■ **AF23** - DSCP 22 (Assured Forwarding Class 2, high drop probability) ■ **AF31** - DSCP 26 (Assured Forwarding Class 3, low drop probability) ■ **AF32** - DSCP 28 (Assured Forwarding Class 3, medium drop probability) ■ **AF33** - DSCP 30 (Assured Forwarding Class 3, high drop probability) ■ **AF41** - DSCP 34 (Assured Forwarding Class 4, low drop probability) ■ **AF42** - DSCP 36 (Assured Forwarding Class 4, medium drop probability) ■ **AF43** - DSCP 38 (Assured Forwarding Class 4, high drop probability) ■ **CS0** - DSCP 0 (Class Selector 0: Default) ■ **CS1** - DSCP 8 (Class Selector 1: Scavenger) ■ **CS2** - DSCP 16 (Class Selector 2: OAM) ■ **CS3** - DSCP 24 (Class Selector 3: Signaling) ■ **CS4** - DSCP 32 (Class Selector 4: Real time) ■ **CS5** - DSCP 40 (Class Selector 5: Broadcast video) ■ **CS6** - DSCP 48 (Class Selector 6: Network control) ■ **CS7** - DSCP 56 (Class Selector 7) ■ **EF** - DSCP 46 (Expedited Forwarding)
`ecn` *<ECN-VALUE>*	Specifies an Explicit Congestion Notification value. Range: 0- 3.
`ip-precedence` *<IP-PRECEDENCE-VALUE>*	Specifies an IP precedence value. Range: 0-7.
`tos` *<TOS-VALUE>*	Specifies the Type of Service value. Range: 0-31.
`fragment`	Not supported.
`vlan` *<VLAN-ID>*	Not supported.
`ttl` *<TTL-VALUE>*	Not supported.
`count`	Keeps the hit counts of the number of packets matching this ACE.

Parameter	Description
`log`	Keeps a log of the number of packets matching this ACE. Works with `deny` actions but not with `permit` actions. Works with ACLs applied on ingress, egress, or Control Plane.
`[<SEQUENCE-NUMBER>] comment <TEXT-STRING>`	Adds a comment to an ACE. The **no** form removes only the comment from the ACE.

Usage

- If the ***<IP-PROTOCOL-NUM>*** parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- The HPE Aruba Networking 8400 Switch series cannot match on MLD (Multicast Listener Discovery) packets using the ICMPv6 protocol number 58 due to the Hop-by-Hop options extension header which is present on MLD packets. The presence of the extension header in the packet causes the 8400 to match on protocol number 0 (HOPOPT) and not protocol number 58 (ICMPv6). Therefore, to achieve the wanted matchings, specify protocol 0 (NOT 58) in your ACEs.
  For example, to permit the joins from `ff12::01` using protocol 0 in the ACE:
  `10 permit 0 any ff12::0/64`
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating an IPv6 ACL with four entries:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 10 permit udp any 2001::1/64
switch(config-acl-ipv6)# 20 permit tcp 2001:2001::2:1/128 gt 1023 any
switch(config-acl-ipv6)# 30 permit tcp 2001:2011::1/64 any
switch(config-acl-ipv6)# 40 deny any any any count
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
        10 permit                          udp
           any
           2001::1/64
        20 permit                          tcp
           2001:2001::2:1                   >  1023
           any
        30 permit                          tcp
           2001:2011::1/64
```

```
          any
       40 deny                            any
          any
          any
          Hit-counts: enabled
```

Adding a comment to an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
        10 permit                         udp
           any
           2001::1/64
        20 Permit all TCP ephemeral ports
           permit                         tcp
           2001:2001::2:1                  >  1023
           any
        30 permit                         tcp
           2001:2011::1/64
           any
        40 deny                           any
           any
           any
           Hit-counts: enabled
```

Removing a comment from an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 20 comment
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
        10 permit                         udp
           any
           2001::1/64
        20 permit                         tcp
           2001:2001::2:1                  >  1023
           any
        30 permit                         tcp
           2001:2011::1/64
```

```
         any
      40 deny                            any
         any
         any
         Hit-counts: enabled
```

Adding an ACE to an existing IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::1/64 any
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
        10 permit                          udp
           any
           2001::1/64
        20 permit                          tcp
           2001:2001::2:1                   >  1023
           any
        25 permit                          icmpv6
           2001::1/64
           any
        30 permit                          tcp
           2001:2011::1/64
           any
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Replacing an ACE in an existing IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::2:1/64 any
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
        10 permit                          udp
           any
           2001::1/64
        20 permit                          tcp
           2001:2001::2:1                   >  1023
           any
```

```
       25 permit                          icmpv6
          2001::2:1/64
          any
       30 permit                          tcp
          2001:2011::1/64
          any
       40 deny                            any
          any
          any
          Hit-counts: enabled

Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
       10 permit                          udp
          any
          2001::1/64
       20 permit                          tcp
          2001:2001::2:1                   >  1023
          any
       25 permit                          icmpv6
          2001::2:1/64
          any
       30 permit                          tcp
          2001:2011::1/64
          any
       40 deny                            any
          any
          any
          Hit-counts: enabled
```

Removing an ACE from an IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 25
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
       10 permit                          udp
          any
          2001::1/64
       20 permit                          tcp
          2001:2001::2:1                   >  1023
          any
       30 permit                          tcp
          2001:2011::1/64
          any
       40 deny                            any
```

```
            any
            any
            Hit-counts: enabled

Type        Name
  Sequence Comment
            Action                          L3 Protocol
            Source IP Address               Source L4 Port(s)
            Destination IP Address          Destination L4 Port(s)
            Additional Parameters
-------------------------------------------------------------------------------
IPv6        MY_IPV6_ACL
        10 permit                           udp
            any
            2001::1/64
        20 permit                           tcp
            2001:2001::2:1                   >  1023
            any
        30 permit                           tcp
            2001:2011::1/64
            any
        40 deny                             any
            any
            any
            Hit-counts: enabled
```

Removing an IPv6 ACL:

```
switch(config)# no access-list ipv6 MY_IPV6_ACL

switch(config)# do show access-list
Type        Name
  Sequence Comment
            Action                          L3 Protocol
            Source IP Address               Source L4 Port(s)
            Destination IP Address          Destination L4 Port(s)
            Additional Parameters
-------------------------------------------------------------------------------
IPv6        MY_IPV6_ACL2
         1 permit                           udp
            any
            2001::1/64
         2 Permit all TCP ephemeral ports
            permit                          tcp
            2001:2001::2:1                   >  1023
            any
         3 permit                           tcp
            2001:2011::1/64
            any
         4 deny                             any
            any
            any
            Hit-counts: enabled
```

Command History

Release	Modification
10.12	Allow ACLs applied to the Control Plane to be logged.

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **access-list ipv6 <ACL-NAME>** command takes you into the named ACL context where you enter the ACEs.	Administrators or local user group members with execution rights for this command.

access-list log-timer

```
access-list log-timer {default|<INTERVAL>}
```

Description

Sets the log timer interval for all ACEs that have the **log** parameter configured.

Parameter	Description
`default`	Resets the log timer to its default 300 seconds.
`<INTERVAL>`	Specifies the log timer interval in seconds. Range: 5 to 300.

Usage

- ACL logging keeps a log of the number of packets matching this ACE. Works with `deny` actions but not with `permit` actions. Works with ACLs applied on ingress, egress, or Control Plane.
- The first packet that matches an ACE with the **log** parameter within an ACL log timer window (configured with the **access-list log-timer** command) has its header contents extracted and sent to the configured logging destination, such as the console and syslog server. Each time the ACL log timer expires, a summary of all ACEs with **log** configured are sent to the logging destination. This capability allows throttling of logging ACL hits.
- If no further log messages are generated in the wait-period, the switch suspends the timer and resets itself to log as soon as a new match occurs.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with the `log` option is logged. Any packets, matching other ACL types, do not create a log until the log-timer wait-period is over. At the end of the wait-period, a summary log is made of all the ACLs that were matched, regardless of type.
- You may see a minor discrepancy between the ACL logging statistics and the hit counts statistics due to the time required to record the log message.

Examples

Although these examples use debug logging, you can alternatively use event logging.

Enabling debug logging for the ACL logging module:

```
switch# debug acl log severity info
switch# show debug
-----------------------------------------------------------------
module sub_module severity vlan  port   ip     mac  instance vrf
-----------------------------------------------------------------
acl    acl_log    info     ----- -----  -----  ---- -----    ---
```

Setting the debug destination to console with the minimum security level of info:

```
switch# debug destination console severity info
switch# show debug destination
----------------------------------------------------------------------
                show debug destination
```

```
--------------------------------------------------------------------
CONSOLE:info
```

Setting the access list log-timer to 30 seconds:

```
switch(config)# access-list log-timer 30
switch(config)# do show access-list log-timer
ACL log timer length (frequency): 30 seconds
```

Creating an IPv4 ACL with one entry with the log parameter:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# deny icmp 1.1.1.1 1.1.1.2 log
switch(config-acl-ip)# do show access-list
Type        Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_ACL
        10 deny                            icmp
           1.1.1.1
           1.1.1.2
           Logging: enabled
           Hit-counts: enabled
```

Enabling interface 1/1/1 and applying the ACL:

```
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# apply access-list ip MY_IP_ACL in
switch(config-if)# do show running-config interface 1/1/1
interface 1/1/1
   no shutdown
   apply access-list ip MY_IP_ACL in
   no routing
   vlan access 1
   exit
```

Sending packets that will match the ACE and observe the ACL logging message on the console:

```
2017-10-10T20:13:36.044+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_
LOG|
List MY_IP_ACL, seq# 10 denied icmp 1.1.1.1 -> 1.1.1.2 type 8 code 0,
on vlan 1, port 1/1/1, direction in
```

When the access list log-timer expires, the summary message is printed on the console. The number 30 is the number of packets received during the last access list log-timer window.

```
2017-10-10T20:14:06.051+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_
LOG|
MY_IP_ACL on 1/1/1 (in): 30  10 deny icmp 1.1.1.1 1.1.1.2 log count
```

Resetting the ACL log timer to the default value:

```
switch(config)# access-list log-timer default
```

Command History

Release	Modification
10.12	Allow ACLs applied to the Control Plane to be logged.
10.09	<INTERVAL> parameter range changed to **5 to 300**. Was 30 to 300.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

access-list mac

```
access-list mac <ACL-NAME>
no access-list mac <ACL-NAME>

[<SEQUENCE-NUMBER>]
{permit|deny}
{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>}]}
{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>}]}
{any|aarp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|
      ipx-arpa|ipx-non-arpa|is-is|lldp|mpls-multicast|mpls-unicast|q-in-q|
      rbridge|trill|wake-on-lan|<NUMERIC-ETHERTYPE>}
[pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment
```

Description

Creates a MAC Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence numbers. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><ACL-NAME></td><td>Specifies the name of this ACL.</td></tr>
<tr><td><SEQUENCE-NUMBER></td><td>Specifies a sequence number for the ACE. Range: 1 to 4294967295.</td></tr>
<tr><td>{permit|deny}</td><td>Specifies whether to permit or deny traffic matching this ACE.</td></tr>
<tr><td>comment</td><td>Specifies storing the remaining entered text as an ACE comment.</td></tr>
<tr><td>{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>}]}</td><td>Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword any. You can optionally include the following:
<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).</td></tr>
<tr><td>{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>}]}</td><td>Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword any. You can optionally include the following:
<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).</td></tr>
<tr><td>{any|aarp|appletalk| ... |wake-on-lan|<NUMERIC-ETHERTYPE></td><td>Specifics the protocol encapsulated in the Ethernet frame. The encapsulated protocol is identified by the EtherType Ethernet field. The EtherType is specified in one of the following three ways:
■ any - any EtherType.
■ <NUMERIC-ETHERTYPE> - the numerical EtherType protocol number. Range: 0x600 to 0xffff.
■ One of these EtherType protocol name keywords:
◦ aarp
◦ appletalk
◦ arp
◦ fcoe
◦ fcoe-init
◦ ip
◦ ipv6
◦ ipx-arpa
◦ ipx-non-arpa
◦ is-is
◦ lldp
◦ mpls-multicast
◦ mpls-unicast
◦ q-in-q
◦ rbridge
◦ trill
◦ wake-on-lan</td></tr>
<tr><td>pcp <PCP-VALUE></td><td>Specifies 802.1Q QoS Priority Code Point value. Range: 0 to 7.</td></tr>
<tr><td>vlan <VID></td><td>Specifies a VLAN ID. The VLAN ID must exist.</td></tr>
</table>

Parameter	Description
	NOTE: This parameter cannot be used in any ACL that will be applied to a VLAN.
`count`	Keeps the hit counts of the number of packets matching this ACE.
`log`	Keeps a log of the number of packets matching this ACE. Works with `deny` actions but not with `permit` actions. Works with ACLs applied on ingress or egress.

Usage

When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating a MAC ACL with four entries:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
switch(config-acl-ip)# 20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
switch(config-acl-ip)# 30 permit any any appletalk vlan 40
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        10 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
        20 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
        30 permit                          appletalk
           any
           any
           VLAN: 40
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Adding a comment to an existing MAC ACE:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 30 comment Permit all vlan-40 tagged Appletalk traffic
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        10 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
        20 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
        30 Permit all vlan-40 tagged Appletalk traffic
           permit                          appletalk
           any
           any
           VLAN: 40
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Removing a comment from an existing MAC ACE:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-mac)# no 30 comment
switch(config-acl-mac)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        10 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
        20 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
        30 permit                          appletalk
           any
           any
           VLAN: 1
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Adding an ACE to an existing MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xffee
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type        Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        10 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
        20 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
        30 permit                          appletalk
           any
           any
           VLAN: 1
        35 permit                          0xffee
           any
           aabb.cc11.1234
        40 deny                            any
           any
           any
           Hit-counts: enabled
```

Replacing an ACE in an existing MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xeeee
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type        Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        10 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
        20 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
        30 permit                          appletalk
           any
           any
           VLAN: 1
        35 permit                          0xeeee
```

```
          any
          aabb.cc11.1234
       40 deny                           any
          any
          any
          Hit-counts: enabled
```

Removing an ACE from an MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# no 35
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type       Name
  Sequence Comment
          Action                         EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
       10 permit                         ipv6
          1122.3344.5566/ffff.ffff.0000
          any
       20 permit                         any
          aaaa.bbbb.cccc
          1111.2222.3333
          QoS Priority Code Point: 4
       30 permit                         appletalk
          any
          any
          VLAN: 1
       40 deny                           any
          any
          any
          Hit-counts: enabled
```

Removing a MAC ACL:

```
switch(config)# no access-list mac MY_MAC_ACL

switch(config)# do show access-list
Type       Name
  Sequence Comment
          Action                         EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL2
        1 permit                         ipv6
          1122.3344.5566/ffff.ffff.0000
          any
        2 permit                         any
          aaaa.bbbb.cccc
          1111.2222.3333
          QoS Priority Code Point: 4
```

```
         3 Permit all vlan-40 tagged Appletalk traffic
           permit                          appletalk
           any
           any
           VLAN: 1
         4 deny                            any
           any
           any
           Hit-counts: enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The `access-list mac <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.	Administrators or local user group members with execution rights for this command.

access-list resequence

`access-list {ip|ipv6|mac} <ACL-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>`

Description

Resequences the ACE sequence numbers in an ACL.

Parameter	Description
`{ip|ipv6|mac}`	Specifies the ACL type.
`<ACL-NAME>`	Specifies the ACL name.
`<STARTING-SEQUENCE-NUMBER>`	Specifies the starting sequence number.
`<INCREMENT>`	Specifies the sequence number increment.

Examples

Resequencing an IPv4 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ip MY_IP_ACL resequence 1 1
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type        Name
  Sequence Comment
```

```
          Action                          L3 Protocol
          Source IP Address               Source L4 Port(s)
          Destination IP Address          Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv4      MY_IP_ACL
        1 permit                          udp
          any
          172.16.1.0/255.255.255.0
        2 permit                          tcp
          172.16.2.0/255.255.0.0           >  1023
          any
        3 permit                          tcp
          172.26.1.0/255.255.255.0
          any
          dscp: AF11
          ack
          syn
        4 deny                            any
          any
          any
          Hit-counts: enabled
```

Resequencing an IPv6 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ipv6 MY_IPV6_ACL resequence 1 1
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
  Sequence Comment
          Action                          L3 Protocol
          Source IP Address               Source L4 Port(s)
          Destination IP Address          Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv6      MY_IPV6_ACL
        1 permit                          udp
          any
          2001::1/64
        2 Permit all TCP ephemeral ports
          permit                          tcp
          2001:2001::2:1                   >  1023
          any
        3 permit                          tcp
          2001:2011::1/64
          any
        4 deny                            any
          any
          any
          Hit-counts: enabled
Type      Name
  Sequence Comment
          Action                          L3 Protocol
          Source IP Address               Source L4 Port(s)
          Destination IP Address          Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv6      MY_IPV6_ACL
        1 permit                          udp
          any
```

```
            2001::1/64
          2 Permit all TCP ephemeral ports
            permit                          tcp
            2001:2001::2:1                   >  1023
            any
          3 permit                          tcp
            2001:2011::1/64
            any
          4 deny                            any
            any
            any
            Hit-counts: enabled
```

Resequencing a MAC ACL to start at 1 with an increment of 1:

```
switch(config)# access-list mac MY_MAC_ACL resequence 1 1
switch(config-acl-mac)# exit

switch(config)# do show access-list
Type        Name
  Sequence Comment
            Action                          EtherType
            Source MAC Address
            Destination MAC Address
            Additional Parameters
-------------------------------------------------------------------------------
MAC         MY_MAC_ACL
          1 permit                          ipv6
            1122.3344.5566/ffff.ffff.0000
            any
          2 permit                          any
            aaaa.bbbb.cccc
            1111.2222.3333
            QoS Priority Code Point: 4
          3 Permit all vlan-40 tagged Appletalk traffic
            permit                          appletalk
            any
            any
            VLAN: 1
          4 deny                            any
            any
            any
            Hit-counts: enabled
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

access-list reset

```
access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset
```

Description

Changes the user-specified ACL configuration to match the active ACL configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Parameter	Description
`all|ip ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>`	Specifies **one** of the following: ▪ a reset of `all` ACLs. ▪ a reset of a named IPv4 ACL. ▪ a reset of a named IPv6 ACL. ▪ a reset of a named MAC ACL.

Usage

The output of the **show access-list** command displays the active configuration of the product. The active configuration is the ACLs that have been configured and accepted by the system. The output of the **show access-list** command with the **configuration** parameter, displays the ACLs that have been configured. The output of this command may not be the same as what was programmed in hardware or what is active on the product.

If the active ACLs and user-configured ACLs are not the same, a warning message is displayed in the output of the show command. Modify the user-configured ACL until the warning message is no longer displayed or run the **access-list reset** command to change the user-specified configuration to match the active configuration.

Examples

Apply an ACL with TCP acknowledgments (ACKs) on ingress, which is unsupported by hardware:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Displaying the user-specified configuration:

```
switch(config)# do show access-list commands
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list commands configuration
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/255.255.0.0 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
```

```
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
IPv4       TEST_ACL

switch(config)# do show access-list configuration
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
IPv4       TEST_ACL
        10
           permit                          tcp
           172.16.2.0/255.255.0.0
           any
           ack

! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list commands configuration
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/255.255.0.0 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
```

```
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
IPv4       TEST_ACL

switch(config)# do show access-list configuration
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
IPv4       TEST_ACL
        10
           permit                          tcp
           172.16.2.0/255.255.0.0
           any
           ack
```

Resetting the user-specified configuration to match the active configuration.

```
switch(config)# access-list ip TEST_ACL reset
```

Displaying the updated user-specified configuration.

```
switch(config)# do show access-list commands
access-list ip TEST_ACL
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
interface 1/1/1
    apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       TEST_ACL

switch(config)# do show access-list configuration
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
```

```
-------------------------------------------------------------------------------
IPv4       TEST_ACL
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

access-list secure-update

```
access-list secure-update
no access list secure-update
```

Description

This command determines if access lists are updated using the secure-update feature. Secure-update is enabled by default.

When secure update is enabled and an ACL is updated or replaced, one or more override entries are installed in the TCAM table(s) containing the ACL that is being modified. As a result, all traffic of the same type as the currently configured ACL will be denied on the interfaces to which the ACL is applied. This ensures that traffic is not temporarily allowed while modifying an ACL. Upon completion of the update, the TCAM override entries are uninstalled and traffic resumes ACL matching.

The **no** version of this command disables this feature. If secure-update is disabled, there will be no override entry installed. This results in the faster modification of an ACL and ensures that there is no interruption to previously permitted traffic, but may temporarily allow previously denied traffic to pass through the switch. Once the ACL has been modified, traffic will be processed by the updated ACL.

Examples

Disabling secure update:

```
switch(config)# no access-list secure-update
```

Reenabling secure update:

```
switch(config)# access-list secure-update
```

Related Commands

Command	Description
`vsx-sync acl-secure-update`	If this setting is enabled and the primary VSX node has configurations with the access list secure-update feature enabled, this configuration can synchronize to the secondary peer. This setting is disabled by default. Refer to the *VSX Guide* for details.

Command History

Release	Modification
10.13	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

apply access-list control-plane

```
apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>
no apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>
```

Description

Applies an ACL to the specified VRF.

The **no** form of this command removes application of the ACL from the specified VRF.

Parameter	Description
`ip|ipv6`	Specifies the ACL type: `ip` for IPv4, or`ipv6` for IPv6.
<ACL-NAME>	Specifies the ACL name.
`vrf` *<VRF-NAME>*	Specifies the VRF name.

Usage

Only one ACL per type (**ip**, or **ipv6**) may be applied to a Control Plane VRF at a time. Therefore, using the **apply access-list control-plane** command on a VRF with an already-applied ACL of the same type, will replace the applied ACL.

Examples

Applying **My_ip_ACL** to Control Plane traffic on the default VRF:

```
switch(config)# apply access-list ip My_ip_ACL control-plane vrf default
```

Replacing My_ip_ACL with **My_Replacement_ACL** on the default VRF:

```
switch(config)# apply access-list ip My_Replacement_ACL control-plane vrf default
```

Remove (unapply) the **My_Replacement_ACL** from the default VRF. Any other interfaces or VLANs with My_Replacement_ACL applied are unaffected.

```
switch(config)# no apply access-list ip My_Replacement_ACL control-plane vrf
default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

apply access-list (to interface or LAG)

`no apply access-list {ip | ipv6 | mac} <ACL-NAME> {in | out}`

Description

Applies an ACL to the interface (Individual front plane port) or Link Aggregation Group (LAG) identified by the current interface or LAG context.

The **no** form of this command removes application of the ACL from the current interface or LAG identified by the current interface or LAG context.

Parameter	Description
`ip|ipv6|mac`	Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.
<ACL-NAME>	Specifies the ACL name.
in	Selects the inbound (ingress) traffic direction.
`out`	Selects the outbound (egress) traffic direction. Only for IPv4 ACLs applied to route-only ports. Not available for ACLs applied to IPv4 bridged ports, IPv6 ports, or MAC ACLs applied to ports.

Usage

- Each ACL of a given type can be applied to the same interface or LAG once in each direction. Therefore, using the **apply access-list** command on an interface or LAG with an already-applied ACL of the same typewill replace the currently applied ACL.
- An ACL can be applied to an individual front plane port or to a Link Aggregation Group (LAG).

- A port that is a member of a LAG with an applied ACL cannot have a different ACL applied to that member port.
- When the port membership of a LAG with an applied ACL is changed, the LAG ACL is automatically applied or removed from that port depending on the modification type.
- No ACLs (including ACLs for IPv4, IPv6, and MAC) are supported in egress on the Layer 2 interface. Egress ACLs can only be applied to Layer 3 (route-only) interfaces. Applying an egress ACL to a Layer 2 interface results in an error.

Examples

Applying **My_IP_ACL** to ingress traffic on interface range 1/1/10 to 1/1/12:

```
switch(config)# int 1/1/10-1/1/12
switch((config-if-<1/1/10-1/1/12>)# apply access-list ip My_IP_ACL in
switch((config-if-<1/1/10-1/1/12>)# exit
```

Applying **MY_IP_ACL** to ingress traffic on LAG 100 and egress traffic on interface 1/1/2:

```
switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ip MY_IP_ACL in
switch(config-lag-if)# exit

switch(config)# interface 1/1/2
switch(config-if)# apply access-list ip MY_IP_ACL out
switch(config-if)# exit
switch(config)#
```

Applying **MY_IPV6_ACL** to ingress traffic on interface 1/1/1 and to ingress traffic on LAG 100:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-if)# exit

switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-lag-if)# exit
switch(config)#
```

Applying **MY_MAC_ACL** to ingress traffic on interface 1/1/1 and ingress traffic on interface 1/1/2:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit

switch(config)# interface 1/1/2
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit
switch(config)#
```

Replacing **MY_IP_ACL** with **MY_REPLACEMENT_ACL** on interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# apply access-list ip MY_REPLACEMENT_ACL out
```

```
switch(config-if)# exit
switch(config)#
```

Unapplying **MY_REPLACEMENT_ACL** from interface 1/1/2 (out):

```
switch(config)# interface 1/1/2
switch(config-if)# no apply access-list ip MY_REPLACEMENT_ACL out
switch(config-if)# exit
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if` `config-lag-if`	Administrators or local user group members with execution rights for this command.

apply access-list (to interface VLAN)

```
apply access-list {ip|ipv6} <ACL-NAME> routed-in
no apply access-list {ip|ipv6} <ACL-NAME> routed-in
```

Description

Applies an ACL to the interface VLAN (or range of interface VLANs) identified by the current interface VLAN context. Using the apply access-list command on an interface VLAN interface with an already-applied ACL of the same type will replace the currently-applied ACL.

The **no** form of this command removes application of the ACL from the interface VLAN (or range of interface VLANs) identified by the current interface VLAN context.

Parameter	Description
`ip|ipv6`	Specifies the ACL type: **ip** for IPv4, **ipv6** for IPv6.
`<ACL-NAME>`	Specifies the ACL name.
`routed-in`	Selects the routed inbound (routed ingress) traffic direction.

Usage

- Each ACL of a given type can be applied to the same interface VLAN once. Therefore, using the **apply access-list** command on an interface VLAN with an already-applied ACL of the same type, will replace the applied ACL.
- When an ACL is applied to an interface VLAN, it will create hardware entries on all line cards regardless of whether an interface VLAN member exists on any specific line card.

Examples

Creating an IPv4 ACL and applying it to routed ingress traffic on interface VLAN vlan100:

```
switch(config)# access-list ip test
switch(config-acl-ip)# 10 permit any 1.1.1.2 2.2.2.2 count
switch(config-acl-ip)# 20 permit any 1.1.1.2 2.2.2.1 count
switch(config-acl-ip)# 30 permit any 2.2.2.2 1.1.1.2 count
switch(config-acl-ip)# 40 permit any 2.2.2.2 1.1.1.1 count
switch(config-acl-ip)# 50 permit any any any count
switch(config-acl-ip)# exit
switch(config)#
switch(config)# interface vlan100
switch(config-if-vlan)# apply access-list ip test routed-in
```

Applying My_ip_ACL to routed ingress traffic on interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# apply access-list ip My_ip_ACL routed-in
```

Applying My_ipv6_ACL to routed ingress traffic on interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# apply access-list ipv6 My_ip_ACL routed-in
```

Applying My_ip_ACL to routed ingress traffic on interface VLANs 20 to 25:

```
switch(config)# interface vlan 20-25
switch(config-if-vlan-<20-25>)# apply access-list ip My_ip_ACL routed-in
```

Replacing My_ipv6_ACL with My_Replacement_ACL on interface VLAN 10 (following the above examples):

```
switch(config)# interface vlan 10
switch(config-if-vlan)# apply access-list ipv6 My_Replacement_ACL routed-in
```

Removing (unapplying) My_Replacement_ACL on interface VLAN 10. Any other interfaces or VLANs with My_Replacement_ACL applied are not affected:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no apply access-list ipv6 My_Replacement_ACL routed-in
```

Removing (unapplying) My_ip_ACL on interface VLANs 20 to 25. Any other interfaces or VLANs with My_ip_ACL applied are not affected:

```
switch(config)# interface vlan 20-25
switch(config-if-vlan-<20-25>)# no apply access-list ip My_ip_ACL routed-in
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config-if-vlan`	Administrators or local user group members with execution rights for this command.

apply access-list (to VLAN)

```
apply access-list {ip|ipv6|mac} <ACL-NAME> in
no apply access-list {ip|ipv6|mac} <ACL-NAME> in
```

Description

Applies an ACL to the VLAN identified by the current VLAN context.

The **no** form of this command removes application of the ACL from the VLAN identified by the current VLAN context.

Parameter	Description
`ip|ipv6|mac`	Specifies the ACL type: **ip** for IPv4, **ipv6** for IPv6, or **mac** for MAC ACL.
<ACL-NAME>	Specifies the ACL name.
`in`	Selects the inbound (ingress) traffic direction.

Usage

- Each ACL of a given type can be applied to the same VLAN once. Therefore, using the `apply access-list` command on a VLAN with an already-applied ACL of the same type, will replace the applied ACL.
- When an ACL is applied to a VLAN, it will create hardware entries on all line cards regardless of whether a VLAN member exists on any specific line card.

Examples

Applying My_ip_ACL to ingress traffic on VLAN range 20 to 25:

```
switch(config)# vlan 20-25
switch(config-vlan-<20-25>)# apply access-list ip My_ip_ACL in
```

Applying My_ip_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ip My_ip_ACL in
```

Applying My_ipv6_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ipv6 My_ipv6_ACL in
```

Applying My_mac_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list mac My_mac_ACL in
```

Replacing My_ipv6_ACL with My_Replacement_ACL on VLAN 10 (following the preceding examples):

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ipv6 My_Replacement_ACL in
```

Removing (unapplying, Specifies the ACL type: **ip** for IPv4, **ipv6** for IPv6, or **mac** for MAC ACL.) several ACLs on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# no apply access-list ipv6 My_Replacement_ACL in
switch(config-vlan-10)# no apply access-list mac My_mac_ACL in
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-vlan`	Administrators or local user group members with execution rights for this command.

clear access-list hitcounts

```
clear access-list hitcounts { all | [{ip|ipv6|mac} <ACL-NAME>]
                  [interface <IF-NAME>| vlan <VLAN-ID>] [in|out|routed-in] }
```

Description

Clears the hit counts for ACLs with ACEs that include the `count` keyword.

Parameter	Description
`all`	Selects all ACLs.
`ip|ipv6|mac`	Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.
<ACL-NAME>	Specifies the ACL name.
`interface` *<IF-NAME>*	Specifies the interface name (port or LAG).

Parameter	Description
`vlan <VLAN-ID>`	Specifies the VLAN.
`in`	Selects the inbound (ingress) traffic direction.
`out`	(not applicable to VLANs) selects the outbound (egress) traffic direction. Only for IPv4 ACLs applied to route-only ports. Not available for ACLs applied to either IPv4 bridged ports or IPv6 ports, or for MAC ACLs applied to ports or VLANS.
`routed-in`	Selects the routed traffic direction on which the ACL is applied. **NOTE:** This is only available for IPv4 and IPv6 ACLs applied to interface VLANs. **routed-in** selects the routed inbound (routed ingress) traffic direction.

Examples

Clearing the hit counts for My_ip_ACL applied to port 1/1/2 (egress):

```
switch# clear access-list hitcounts ip My_ip_ACL interface 1/1/2 out
```

Clearing the hit counts for My_ip_ACL applied to VLAN 10 (ingress):

```
switch# clear access-list hitcounts ip My_ip_ACL vlan 10 in
```

Clearing the hit counts for all ACLs:

```
switch# clear access-list hitcounts all
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

clear access-list hitcounts control-plane

`clear access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>`

Description

Clears the hit counts for ACLs applied to the Control Plane VRF.

Parameter	Description
`ip|ipv6`	Specifies the ACL type: **ip** for IPv4, or **ipv6** for IPv6.
<ACL-NAME>	Specifies the ACL name.
`vrf` *<VRF-NAME>*	Specifies the VRF name.

Examples

Clearing the hit counts for an IPv4 ACL applied to the Control Plane `default` VRF:

```
switch# clear access-list hitcounts ip My_ipv4_ACL control-plane vrf default
```

Clearing the hit counts for an IPv6 ACL applied to the Control Plane `default` VRF:

```
switch# clear access-list hitcounts ipv6 My_ipv6_ACL control-plane vrf default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

object-group address resequence

`object-group {ip|ipv6} address` *<OBJECT-GROUP-NAME>* `resequence` *<STARTING-SEQUENCE-NUMBER>* *<INCREMENT>*

Description

Reorders the sequence numbers in an address object group.

Parameter	Description
`ip|ipv6`	Specifies the object group IP address type, either `ip` or `ipv6`.
<OBJECT-GROUP-NAME>	Specifies the address object group name.
<STARTING-SEQUENCE-NUMBER>	Specifies the starting sequence number.
<INCREMENT>	Specifies the sequence number increment.

Examples

Resequencing address object group my_ipv4_addr_group to use sequence numbers 5, 10, 15 and so on:

```
switch(config)# object-group address my_ipv4_addr_group resequence 5 5
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

object-group address reset

`object-group {ip|ipv6} address <OBJECT-GROUP-NAME> reset`

Description

Resets the user configuration back to the active configuration. This command takes immediate effect, it is not saved in the user configuration. Use this command if misconfiguration of an address object group has occurred.

Parameter	Description
`ip|ipv6`	Specifies the object group IP address type, either `ip` or `ipv6`.
<OBJECT-GROUP-NAME>	Specifies the address object group name.

Examples

Resetting IPv4 address object group my_ipv4_group:

```
switch(config)# object-group ip address my_ip_group reset
switch(config)#
```

Resetting IPv6 address object group my_ipv6_group:

```
switch(config)# object-group ipv6 address my_ipv6_group reset
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

object-group all reset

`object-group all reset`

Description

Resets the user configuration back to the active configuration for all object types (address and port). This command takes immediate effect, it is not saved in the user configuration. Use this command if misconfiguration of address object groups and port object groups has occurred. Individual address and port object groups can be reset respectively with the **object-group address reset** and **object-group port** reset commands.

Examples

Resetting the user configuration for all object types (address and port):

```
switch(config)# object-group all reset
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

object-group ip address

Syntax to create an IPv4 address object group and enter its context:

```
object-group ip address <OBJECT-GROUP-NAME>
no object-group ip address <OBJECT-GROUP-NAME>
```

Syntax (within the address object-group context) for creating or removing IPv4 address entries :

```
[<SEQUENCE-NUMBER>]  <IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]

  no <SEQUENCE-NUMBER>
```

Description

Creates an IPv4 address object group comprised of one or more address entries. Address groups are used solely as a shorthand way of specifying groups of addresses in the ACEs that make up ACLs. IPv4 address groups can be used only in the **access-list ip** command. Entering **object-group ip address** with an existing address group name, enables you to modify an existing address group.

The **no** form of this command deletes the entire address group or deletes a particular address group entry identified by sequence number.

Parameter	Description
<OBJECT-GROUP-NAME>	Specifies the address object group name.
<SEQUENCE-NUMBER>	Specifies a sequence number for the address entry. Range: 1 to 4294967295. When omitted, a sequence number 10 larger than the current highest sequence number is auto-assigned. Default auto-assigned sequence numbers are 10, 20, 30, and so on.
*<IP-ADDRESS>***[/{***<PREFIX-LENGTH>***|***<SUBNET-MASK>***}]**	Specifies the IPv4 address. ■ ***<IP-ADDRESS>*** - specifies the IPv4 host address. ■ ***<PREFIX-LENGTH>*** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ■ ***<SUBNET-MASK>*** - specifies the address bits to mask (dotted decimal notation).

Examples

Creating an IPv4 address group with two entries:

```
switch(config)# object-group ip address my_ipv4_addr_group
switch(config-addrgroup-ip)# 10 192.168.0.1
switch(config-addrgroup-ip)# 20 192.168.0.2
switch(config-addrgroup-ip)# exit
switch(config)# show object-group
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv4       my_ipv4_addr_group
        10 192.168.0.1
        20 192.168.0.2
```

Adding an entry to an existing IPv4 address group:

```
switch(config)# object-group ip address my_ipv4_addr_group
switch(config-addrgroup-ip)# 30 192.168.0.3
switch(config-addrgroup-ip)# exit
switch(config)# show object-group
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv4       my_ipv4_addr_group
        10 192.168.0.1
        20 192.168.0.2
        30 192.168.0.3
```

Removing an entry (20) from an existing IPv4 address group:

```
switch(config)# object-group ip address my_ipv4_addr_group
switch(config-addrgroup-ip)# no 20
switch(config-addrgroup-ip)# exit
switch(config)# show object-group
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv4       my_ipv4_addr_group
        10 192.168.0.1
        30 192.168.0.3
```

Removing an IPv4 address group:

```
switch(config)# no object-group ip address my_ipv4_addr_group
switch(config)# show object-group
No object group found.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **object-group ip address** command takes you into the named address group context (with prompt **switch (config-addrgroup-ip)#**) where you enter the addresses.	Administrators or local user group members with execution rights for this command.

object-group ipv6 address

Syntax to create an IPv6 address object group and enter its context:

`object-group ipv6 address <OBJECT-GROUP-NAME>`

`no object-group ipv6 address <OBJECT-GROUP-NAME>`

Syntax (within the address object-group context) for creating or removing IPv6 address entries :

```
[<SEQUENCE-NUMBER>]  <IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]

no <SEQUENCE-NUMBER>
```

Description

Creates an IPv6 address object group comprised of one or more address entries. Address groups are used solely as a shorthand way of specifying groups of addresses in the ACEs that make up ACLs. IPv6 address groups can be used only in the `access-list ipv6` command. Entering `object-group ipv6 address` with an existing address group name, enables you to modify an existing address group.

The **no** form of this command deletes the entire address group or deletes a particular address group entry identified by sequence number.

Parameter	Description
<OBJECT-GROUP-NAME>	Specifies the address object group name.
<SEQUENCE-NUMBER>	Specifies a sequence number for the address entry. Range: 1 to 4294967295. When omitted, a sequence number 10 larger than the current highest sequence number is auto-assigned. Default auto-assigned sequence numbers are 10, 20, 30, and so on.
<IP-ADDRESS>[/{*<PREFIX-LENGTH>*|*<SUBNET-MASK>*}]	Specifies the IPv6 address. ■ *<IP-ADDRESS>* - specifies the IPv6 host address. ◦ *<PREFIX-LENGTH>* - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128. ◦ *<SUBNET-MASK>* - specifies the address bits to mask (dotted decimal notation).

Examples

Creating an IPv6 address group with two entries:

```
switch(config)# object-group ipv6 address my_ipv6_addr_group
switch(config-addrgroup-ipv6)# 10 1000::1
switch(config-addrgroup-ipv6)# 20 1000::2
switch(config-addrgroup-ipv6)# exit
switch(config)# show object-group
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv6       my_ipv6_addr_group
        10 1000::1
        20 1000::2
```

Adding an entry to an existing IPv6 address group:

```
switch(config)# object-group ipv6 address my_ipv6_addr_group
switch(config-addrgroup-ipv6)#
switch(config-addrgroup-ipv6)# 30 1000::3
switch(config-addrgroup-ipv6)# exit
switch(config)# show object-group
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv6       my_ipv6_addr_group
        10 1000::1
        20 1000::2
        30 1000::3
```

Removing an entry (20) from an existing IPv6 address group:

```
switch(config)# object-group ipv6 address my_ipv6_addr_group
switch(config-addrgroup-ipv6)# no 20
switch(config-addrgroup-ipv6)# exit
switch(config)# show object-group
Type        Name
  Sequence L4 Port(s)/IP Address
--------------------------------------------------------------------------------
IPv6        my_ipv6_addr_group
        10 1000::1
        30 1000::3
```

Removing an IPv6 address group:

```
switch(config)# no object-group ipv6 address my_ipv6_addr_group
switch(config)# show object-group
No object group found.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **object-group ipv6 address** command takes you into the named address group context (with prompt **switch (config-addrgroup-ipv6)#**) where you enter the addresses.	Administrators or local user group members with execution rights for this command.

object-group port

Syntax to create a Layer 4 port object group and enter its context:

```
object-group port <OBJECT-GROUP-NAME>

no object-group port <OBJECT-GROUP-NAME>
```

Syntax (within the port object-group context) for creating or removing Layer 4 port entries:

```
  [<SEQUENCE-NUMBER>] { {eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT> }

  no <SEQUENCE-NUMBER>
```

Description

Creates a Layer 4 port object group comprised of one or more port entries. Port groups are used solely as a shorthand way of specifying groups of ports in the ACEs that make up ACLs. Layer 4 port groups can be used only in the **access-list ip** and **access-list ipv6** commands. Entering **object-group port** with an existing port group name, enables you to modify an existing port group.

The **no** form of this command deletes the entire port group or deletes a particular port group entry identified by sequence number.

Parameter	Description
<OBJECT-GROUP-NAME>	Specifies the port object group name.
<SEQUENCE-NUMBER>	Specifies a sequence number for the port entry. Range: 1 to 4294967295. When omitted, a sequence number 10 larger than the current highest sequence number is auto-assigned. Default auto-assigned sequence numbers are 10, 20, 30, and so on.
{ {eq|gt|lt} *<PORT>***|range** *<MIN-PORT><MAX-PORT>* **}**	Specifies the port or port range. Port numbers are in the range of 0 to 65535. ■ **eq *<PORT>*** - specifies the Layer 4 port. ■ **gt *<PORT>*** - specifies any Layer 4 port greater than the indicated port. ■ **lt *<PORT>*** - specifies any Layer 4 port less than the indicated port. ■ **range *MIN-PORT> <MAX-PORT>*** - specifies the Layer 4 port range. **NOTE:** When ACLs using ACEs defined with port groups are applied, the same number of hardware resources are consumed as when the ports are specified directly in the ACEs and not in a group. Keep this in mind when creating port groups that include many ports. Although hardware resource consumption is the same, with or without port groups used, it may not be immediately obvious that some port groups that you have defined, include many ports. It is recommended that you name port groups in a manner that reminds you that a group includes many ports.

Examples

Creating a port group with two entries to cover port 80 plus ports 0 through 50:

```
switch(config)# object-group port my_port_group
switch(config-portgroup)# 10 eq 80
switch(config-portgroup)# 20 range 0 50
switch(config-portgroup)# exit
switch(config)# show object-group
Type        Name
  Sequence L4 Port(s)/IP Address
```

```
--------------------------------------------------------------------------------
Port        my_port_group
        10 eq 80
        20 range 0 50
```

Adding an entry for ports greater than 65525 (covers ports 65526 through 65535):

```
switch(config)# object-group port my_port_group
switch(config-portgroup)# 30 gt 65525
switch(config-portgroup)# exit
switch(config)# show object-group
Type        Name
  Sequence L4 Port(s)/IP Address
--------------------------------------------------------------------------------
Port        my_port_group
        10 eq 80
        20 range 0 50
        30 gt 65525
```

Removing an entry (#20) from the port group:

```
switch(config)# object-group port my_port_group
switch(config-portgroup)# no 20
switch(config-portgroup)# exit
switch(config)# show object-group
Type        Name
  Sequence L4 Port(s)/IP Address
--------------------------------------------------------------------------------
Port        my_port_group
        10 eq 80
        30 gt 65525
```

Removing the port group:

```
switch(config)# no object-group port my_port_group
switch(config)# show object-group
No object group found.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **object-group ip port** command takes you into the named port group context (with prompt **switch(config-**	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
	portgroup)#) where you specify the ports.	

object-group port resequence

`object-group port <OBJECT-GROUP-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>`

Description

Reorders the sequence numbers in a port object group.

Parameter	Description
<OBJECT-GROUP-NAME>	Specifies the port object group name.
<STARTING-SEQUENCE-NUMBER>	Specifies the starting sequence number.
<INCREMENT>	Specifies the sequence number increment.

Examples

Resequencing port object group my_port_group to use sequence numbers 110, 120, 130 and so on:

```
switch(config)# object-group port my_port_group resequence 110 10
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

object-group port reset

`object-group port <OBJECT-GROUP-NAME> reset`

Description

Resets the user configuration back to the active configuration. This command takes immediate effect, it is not saved in the user configuration. Use this command if misconfiguration of a port object group has occurred.

Parameter	Description
<OBJECT-GROUP-NAME>	Specifies the port object group name.

Examples

Resetting port object group my_port_group:

```
switch(config)# object-group port my_port_group reset
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show access-list

Syntax that filters by ACLs applied to an interface, VLAN, or VNI:

```
show access-list [interface <IF-NAME>|vlan <VLAN-ID>] [ip|ipv6|mac]
                 [in|out|routed-in] [commands] [configuration] [vsx-peer]
```

Syntax that filters by the named ACL:

```
show access-list [ip|ipv6|mac] [<ACL-NAME>] [commands] [configuration] [vsx-peer]
```

Description

Shows information about your defined ACLs and where they have been applied. When **show access-list** is entered without parameters, information for all ACLs is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL.
- All ACLs of a specific type.
- The ACL applied to a specific interface (port or split port or LAG).
- The ACL applied to a specific VLAN.
- All IPv4 or IPv6 ACLs applied to interface VLANs (routed in).

Parameter	Description
interface <IF-NAME>	Specifies the interface name (port or LAG).
vlan <VLAN-ID>	Specifies the VLAN.

Parameter	Description
`ip|ipv6|mac`	Specifies the ACL type: ■ **ip** for IPv4, ■ **ipv6** for IPv6, or ■ **mac** for MAC.
`in`	Selects the inbound (ingress) traffic direction.
`out`	(not applicable to VLANs) Selects the outbound (egress) traffic direction. Only for IPv4 ACLs applied to route-only ports. Not available for ACLs applied to either IPv4 bridged ports or IPv6 ports, or for MAC ACLs applied to ports or VLANS.
`routed-in`	Selects the routed traffic direction on which the ACL is applied. **NOTE:** This is only available for IPv4 and IPv6 ACLs applied to interface VLANs. **routed-in** selects the routed inbound (routed ingress) traffic direction.
`<ACL-NAME>`	Specifies the ACL name.
`commands`	Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.
`configuration`	Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Creating an IPv4 ACL, applying it to an interface VLAN (routed in), and then showing ACL information filtered for that interface VAN:

```
switch(config)# access-list ip test
switch(config-acl-ip)# 10 permit any 1.1.1.2 2.2.2.2 count
switch(config-acl-ip)# 20 permit any 1.1.1.2 2.2.2.1 count
switch(config-acl-ip)# 30 permit any 2.2.2.2 1.1.1.2 count
switch(config-acl-ip)# 40 permit any 2.2.2.2 1.1.1.1 count
switch(config-acl-ip)# 50 permit any any any count
switch(config-acl-ip)# exit
switch(config)#
switch(config)# interface vlan100
switch(config-if-vlan)# apply access-list ip test routed-in
switch(config-if-vlan)# exit
switch(config)# show access-list interface vlan100 ip routed-in

Direction
Type       Name
```

```
  Sequence Comment
           Ac  L3 Protocol
           Source IP Address                Source L4 Port(s)
           Destination IP Address           Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
Routed Inbound
IPv4       test
        10
           permit                           any
           1.1.1.2
           2.2.2.2
           Hit-counts: enabled
        20
           permit                           any
           1.1.1.2
           2.2.2.1
           Hit-counts: enabled
        30
           permit                           any
           2.2.2.2
           1.1.1.2
           Hit-counts: enabled
        40
           permit                           any
           2.2.2.2
           1.1.1.1
           Hit-counts: enabled
        50
           permit                           any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
```

Showing an IPv4 ACL:

```
switch# show access-list ip MY_ACL
Type       Name
  Sequence Comment
           Action                           L3 Protocol
           Source IP Address                Source L4 Port(s)
           Destination IP Address           Destination L4 Port(s)
           Additional Parameters
------------------------------------------------------------------------------
IPv4       MY_ACL
        10 permit                           udp
           any
           172.16.1.0/255.255.255.0
        20 permit                           tcp
           172.16.2.0/255.255.0.0           >  1023
           any
        30 permit                           tcp
           172.26.1.0//255.255.255.0
           any
           syn
           ack
           dscp 10
        40 deny                             any
           any
           any
```

```
          Hit-counts: enabled
-------------------------------------------------------------------------------
```

Showing an IPv4 ACL as commands:

```
switch# show access-list ip MY_ACL commands
access-list ip MY_ACL
    10 permit udp any 172.16.1.0/255.255.255.0
    20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
    30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
    40 deny any any any count
```

Showing IPv4 ACLs applied to VLAN 10, inbound:

```
switch# show access-list vlan 10 ip in
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       My_ip_ACL
        10 permit                          udp
           any
           172.16.1.0/255.255.255.0
        20 permit                          tcp
           172.16.2.0/255.255.0.0          >  1023
           any
        30 permit                          tcp
           172.26.1.0//255.255.255.0
           any
           syn
           ack
           dscp 10
        40 deny                            any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
```

Showing IPv6 ACLs applied to LAG 128, inbound:

```
switch# show access-list interface lag128 ipv6 in
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_ACL
        10 permit                          udp
           any
           2001::1/64
        20 permit                          tcp
           2001:2001::2:1/128               >  1023
```

```
          any
       30 permit                         tcp
          2001:2011::1/64
       40 deny                           any
          any
          any
          Hit-counts: enabled
-------------------------------------------------------------------------------
```

Showing an IPv6 ACL as commands:

```
switch# show access-list ipv6 MY_IPV6_ACL commands
access-list ipv6 MY_IPV6_ACL
    10 permit udp any 2001::1/64
    20 permit tcp 2001:2001::2:1/128 gt 1023 any
    40 deny any any any count
```

Showing a MAC ACL:

```
switch# show access-list mac MY_MAC_ACL
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_ACL
        10 permit                          ipv6
           1122.3344.5566/ffff.ffff.0000
           any
        20 permit                          any
           aaaa.bbbb.cccc
           1111.2222.3333
           QoS Priority Code Point: 4
        30 deny                            any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
```

Showing a MAC ACL as commands:

```
switch# show access-list mac MY_MAC_ACL commands
access-list mac MY_MAC_ACL
    10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
    20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
    30 deny any any any count
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show access-list control-plane

```
show access-list [ip|ipv6] [<ACL-NAME>] control-plane [vrf <VRF-NAME>]
                 [commands] [configuration][vsx-peer]
```

Description

Shows information about your defined ACLs that have been applied to the Control Plane. When **show access-list control-plane** is entered without parameters, information for all ACLs applied to the Control Plane is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL that has been applied to the Control Plane.
- All ACLs of a specific type that have been applied to the Control Plane.
- All ACLs applied to the Control Plane for a specific VRF.

Parameter	Description
`ip|ipv6`	Specifies the ACL type: `ip` for IPv4, or`ipv6` for IPv6.
<ACL-NAME>	Specifies the ACL name.
`vrf` *<VRF-NAME>*	Specifies the VRF name.
`[commands]`	Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.
`[configuration]`	Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing an IPv4 ACL applied to the Control Plane `default` VRF:

```
switch# show access-list ip My_ipv4_ACL control-plane vrf default
Type        Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       My_ipv4_ACL
```

```
        10 permit                          udp
           any
           172.16.1.0/24
        20 permit                          tcp
           172.16.2.0/16                    >  1023
           any
        30 permit                          tcp
           172.26.1.0/24
           any
           syn
           ack
           dscp 10
        40 deny                            any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
```

Showing an IPv6 ACL applied to the Control Plane `default` VRF:

```
switch# show access-list ipv6 My_ipv6_ACL control-plane vrf default
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       My_ipv6_ACL
        10 permit                          udp
           any
           2001::1/64
        20 permit                          tcp
           2001:2001::2:1/128               >  1023
           any
        30 permit                          tcp
           2001:2011::1/64
        40 deny                            any
           any
           any
           Hit-counts: enabled
-------------------------------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show access-list hitcounts

```
show access-list hitcounts { [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME> |
                  vlan <VLAN-ID>] [in|out|routed-in] [vsx-peer] }
```

Description

Shows the hit count of the number of times an ACL has matched a packet or frame for ACEs with the **count** keyword. For ACEs without the **count** keyword, a dash is shown in place of a hit count.

Parameter	Description
`ip|ipv6|mac`	Specifies the ACL type: **ip** for IPv4, **ipv6** for IPv6, or **mac** for MAC.
`<ACL-NAME>`	Specifies the ACL name.
`interface <IF-NAME>`	Specifies the interface name (port or split port or LAG).
`vlan <VLAN-ID>`	Specifies the VLAN.
`in`	Selects the inbound (ingress) traffic direction.
`out`	(Not applicable to VLANs) Selects the outbound (egress) traffic direction. Only for IPv4 ACLs applied to route-only ports. Not available for ACLs applied to either IPv4 bridged ports or IPv6 ports, or for MAC ACLs applied to ports or VLANS.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

- ACL hit counts are aggregated across all:
  - Physical interfaces to which the ACL is applied to on ingress.
  - Physical interfaces to which the ACL is applied to on egress.
  - VLANs to which the ACL is applied to on ingress.
  - Interface VLANs to which the IPv4 or IPv6 ACL is applied on routed ingress.
- If an ACL with an ACE with the **count** keyword is applied to multiple physical interfaces or VLANs, the hit counts are aggregated. There is one aggregation for physical interfaces and another for VLANs.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for **My_ip_ACL** applied to port 1/1/2:

```
switch# show access-list hitcounts ip My_ip_ACL interface 1/1/2
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/1-1/1/2,lag1 (out):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
```

Showing the hit counts for **My_ip_ACL** applied to VLAN 10:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
vlan 10,20-100,300 (in):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
```

Showing the hit counts for **My_ip_ACL** applied to interface VLAN 10:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
interface vlan 10,20,30 (routed-in):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
```

Showing the hit counts for My_ip_ACL applied on any interface and direction:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
switch# show access-list hitcounts ip My_ip_ACL
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/1 (in):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count

interface 1/1/1-1/1/2,lag1 (out):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count

interface 1/1/4.1,1/1/10.10 (in):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count

interface 1/1/4.1 (out):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count

interface vlan 10,20,30 (routed-in):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
```

```
interface vlan 80-85 (routed-out):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count

vlan 10,20-100,300 (in):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
vlan 2-5 (out):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count

vrf blue,default,red (control-plane):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
```

Command History

Release	Modification
10.07 or earlier	Updated command output to use interface and VLAN ranges to reflect aggregation.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show access-list hitcounts control-plane

`show access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME> [vsx-peer]`

Description

Shows the hit count of the number of times an ACL (applied to the Control Plane) has matched a packet for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

Parameter	Description
`ip|ipv6`	Specifies the ACL type: **ip** for IPv4, or **ipv6** for IPv6.

Parameter	Description
<ACL-NAME>	Specifies the ACL name.
vrf *<VRF-NAME>*	Specifies the VRF name.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

- ACL hit counts are aggregated across all VRFs to which the ACL is applied to on ingress.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for an IPv4 ACL applied to the Control Plane default VRF:

```
switch# show access-list hitcounts ip My_ipv4_ACL control-plane vrf default
Statistics for ACL My_ip_ACL (ipv4):
vrf default (control-plane):
     Matched Packets  Configuration
                   -  10 permit udp any 172.16.1.0/255.255.255.0
                   0  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                   -  30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11 ack syn
                   0  implicit deny any any any count
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show access-list secure-update

`show access-list secure-update`

Description

Use this command to determine if access lists are updated using the secure-update feature. Secure update is enabled by default.

Examples

Displaying the status of the access list secure-update feature when that feature is enabled:

```
switch(config)# show access-list secure-update
Access-list secure-update is enabled
```

Displaying the status of the access list secure-update feature when that feature is disabled:

```
switch(config)# show access-list secure-update
Access-list secure-update is disabled
```

Related Commands

Command	Description
`access-list secure-update`	This command determines if access lists are updated using the secure-update feature. Secure-update is enabled by default.

Command History

Release	Modification
10.13	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

show capacities

`show capacities <FEATURE> [vsx-peer]`

Description

Shows system capacities and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies a feature. For example, **aaa** or **vrrp**.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing all available capacities for BGP:

```
switch# show capacities bgp

System Capacities: Filter BGP
Capacities Name                                                              Value
----------------------------------------------------------------------------------
-
Maximum number of AS numbers in as-path attribute
32
...
```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring

System Capacities: Filter Mirroring
Capacities Name                                                              Value
----------------------------------------------------------------------------------
-
Maximum number of Mirror Sessions configurable in a system
4
Maximum number of enabled Mirror Sessions in a system
4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                                              Value
----------------------------------------------------------------------------------
-
Maximum number of mstp instances configurable in a system
64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                                              Value
----------------------------------------------------------------------------------
-
Maximum number of VLANs supported in the system
4094      /switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                                              Value
----------------------------------------------------------------------------------
-
Maximum number of VLANs supported in the system
4094
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities-status

```
show capacities-status <FEATURE> [vsx-peer]
```

Description

Shows system capacities status and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies the feature, for example **aaa** or **vrrp** for which to display capacities, values, and status. Required.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

The number of IP neighbors (IPv4+IPv6) entries section in the output of the s**how capacities-status l3-resources** command lists the maximum number of neighbors supported by the platform (with both IPv4 and IPv6 neighbors combined). On HPE Aruba Networking 8320, 8325, 8400, 9300/9300S and 10000 Switch series, IPv6 neighbors will count a value of two for each neighbor, as they take up twice the space in hardware as compared to IPv4 neighbors. In previous releases, the output the **show capacities-status l3-resources** command counted IPv4 and IPv6 neighbors as consuming the same number of resources/space in hardware.

For Spine and L3-core profiles on HPE Aruba Networking 8320, 8325 and 10000 Switch series, the IPv6 Route entry size is still one for each neighbor. For Leaf and L3-agg profiles, the entry size is also one for IPv6 routes with a prefix greater than 64.

Examples

Showing the system capacities status for all features:

```
                   switch# show capacities-status

System Capacities Status
Capacities Status Name                                                        Value
Maximum
---------------------------------------------------------------------------------
-------
Number of active gateway mac addresses in a system                               0
```

```
    16
Number of aspath-lists configured                                              0
    64
Number of community-lists configured                                           0
    64
...
```

Showing the system capacities status for BGP:

```
switch# show capacities-status bgp

System Capacities Status: Filter BGP
Capacities Status Name                                               Value  Maximum
-----------------------------------------------------------------------------------
--
Number of aspath-lists configured                                    0      64
Number of community-lists configured                                 0      64
Number of neighbors configured across all VRFs                       0      50
Number of peer groups configured across all VRFs                     0      25
Number of prefix-lists configured                                    0      64
Number of route-maps configured                                      0      64
Number of routes in BGP RIB                                          0      256000
Number of route reflector clients configured across all VRFs         0      16
```

Showing the system capacities status for L3:

```
switch# show capacities-status l3

System Capacities Status: Filter L3 resources
Capacities Status Name                                                  Value
Maximum
-----------------------------------------------------------------------------------
--
Number of IP neighbor (IPv4+IPv6) entries                                4
49152
Number of IP Directed Broadcast neighbor entries                         0
4096
Number of IPv6 Long Prefix Routes currently configured                   3
5000
Number of IPv6 neighbor(ND) entries                                      4
49152
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured 1
2047
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
      Tunnels currently configured                                       4
2045
Number of routes (IPv4+IPv6) currently configured                        5
65536
Number of IPv4 routes currently configured                               0
65536
Number of IPv6 routes currently configured with prefix 0-64              4
13312
Number of IPv6 routes currently configured with prefix 65-127            2
510
```

```
switch# show capacities-status l3
```

```
System Capacities Status: Filter L3 resources
Capacities Status Name                                                      Value
Maximum
----------------------------------------------------------------------------------
--
Number of IP neighbor (IPv4+IPv6) entries                                    4
49152
Number of IP Directed Broadcast neighbor entries                             0
4096
Number of IPv6 Long Prefix Routes currently configured                       3
5000
Number of IPv6 neighbor(ND) entries                                          4
49152
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured 1
2047
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
      Tunnels currently configured                                           4
2045
Number of routes (IPv4+IPv6) currently configured                            5
65536
Number of IPv4 routes currently configured                                   0
65536
Number of IPv6 routes currently configured with prefix 0-64                  4
13312
Number of IPv6 routes currently configured with prefix 65-127                2
510
```

Command History

Release	Modification
10.13	Updated to show newly supported configuration of IPv6 routes on the ASIC.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show object-group

`show object-group [{ip|ipv6} address | port] [<OBJECT-GROUP-NAME>] [commands] [configuration]`

Description

Shows information about your defined object groups. When **show object-group** is entered without parameters, information for all object groups is shown. The parameters filter the list of object groups for which information is shown.

Parameter	Description
`[{ip|ipv6} address	port]`	Specifies the object group type, either **address** for an IP address, or **port**.
<OBJECT-GROUP-NAME>	Specifies the object group name.
`[commands]`	Specifies that the object group definition is to be shown as the commands and parameters used to create it rather than in tabular form.
`[configuration]`	Specifies that the user-configured object groups be shown as configured. The output of the command with this parameter may not be the same as what is active on the switch due to a misconfigured object group. See *Examples* in this topic.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing configured object groups:

```
switch# show object-group
Type        Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv4        my_address_group
        10 192.168.0.1
        20 192.168.0.3
Port        my_port_group
        10 eq 80
        20 gt 65525
switch#
switch# show object-group commands
object-group ip address my_address_group
    10 192.168.0.1
    20 192.168.0.3
object-group port my_port_group
    10 eq 80
    20 gt 65525
```

Showing a misconfigured object group:

```
switch# show object-group
Type        Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
IPv4        my_address_group
switch#
switch#
Type        Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
```

```
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
IPv4       my_address_group
switch#
switch# show object-group configuration
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
IPv4       my_address_group
        10 192.168.0.1
        20 192.168.0.3
switch#
switch# show object-group commands
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
switch#
switch# show object-group commands configuration
! object-group ip address My_ip_object_group user configuration does not match
! the active hardware configuration. Run 'object-group ip address NAME reset'
! to reset the object group to match the active hardware configuration.
object-group ip address my_address_group
    10 192.168.0.1
    20 192.168.0.3
```

Resetting a misconfigured object group:

```
switch(config)# object-group all reset
switch(config)# exit
switch# show object-group
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv4       my_address_group
switch#
switch# show object-group configuration
Type       Name
  Sequence L4 Port(s)/IP Address
-------------------------------------------------------------------------------
IPv4       my_address_group
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

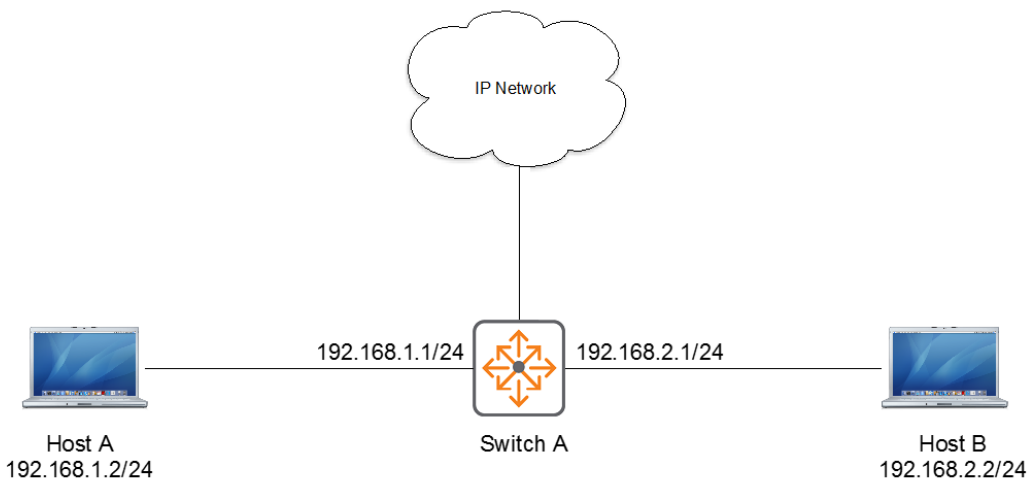
ACL configuration examples

This chapter shows examples for defining and applying IPv4 and IPv6 ACLs.

IPv4 ACL example overview

This example:

- Defines and applies an ACL to interface 1/1/1 on Switch A (see image in this topic) so that Host A is not able to send traffic to Host B, but it can communicate with all other devices in the network.
- Counts blocked packets.



Defining and applying an IPv4 ACL

Procedure

1. Begin defining an IPv4 ACL named FILTER_TO_HOST_B:
   ```
   switch(config)# access-list ip FILTER_TO_HOST_B
   ```
2. Add an ACE that denies access from IP address 192.168.1.2 (Host A) to 192.168.2.2 (Host B):
   ```
   switch(config-acl-ip)# deny any 192.168.1.2 192.168.2.2 log
   ```
3. Add an ACE that allows access from all other IP addresses:
   ```
   switch(config-acl-ip)# permit any any any
   ```
4. Exit the ACL definition:
   ```
   switch(config-acl-ip)# exit
   ```
5. Enter the context of the interface to which you will apply the ACL:
   ```
   switch(config)# interface 1/1/1
   ```
6. Apply the FILTER_TO_HOST_B ACL to inbound (ingress) traffic:
   ```
   switch(config-if)# apply access-list ip FILTER_TO_HOST_B in
   ```

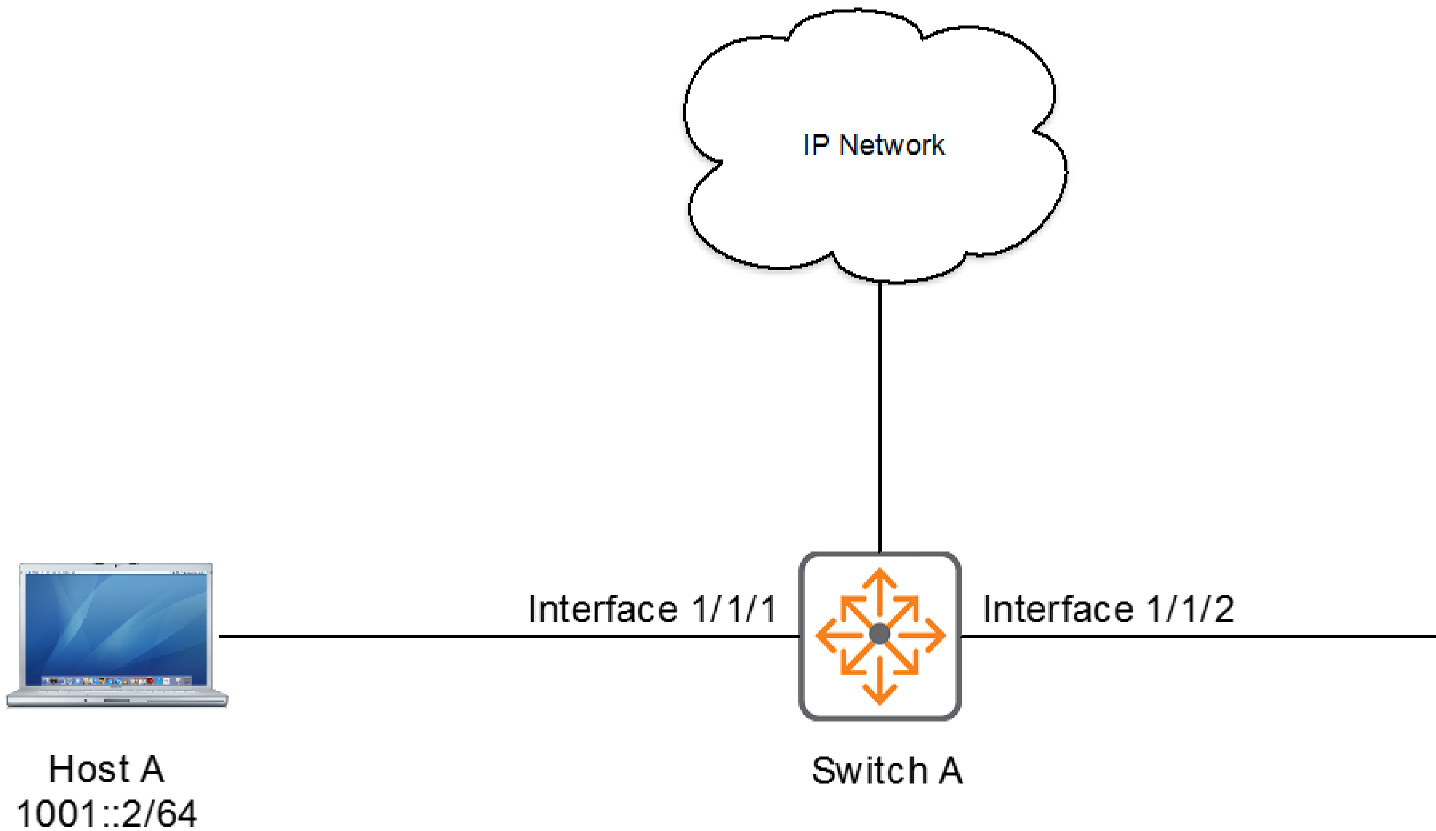
7. Show your ACL:

```
switch(config-if)# exit
switch# show access-list ip FILTER_TO_HOST_B
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-----------------------------------------------------------------------------
---
IPv4       FILTER_TO_HOST_B
        10
           deny                            any
           192.168.1.2
           192.168.2.2
           Logging: enabled
           Hit-counts: enabled
        20
           permit                          any
           any
           any
-----------------------------------------------------------------------------
---
```

IPv6 ACL example overview

This example:

- Defines and applies an ACL to interface 1/1/1 on Switch A (see image in this topic) so that Host A is not able to send traffic to Host B, but it can communicate with all other devices in the network.
- Counts blocked packets.



Defining and applying an IPv6 ACL

Procedure

1. Begin defining an IPv6 ACL named V6_INPUT_FILTER:

   `switch(config)# access-list ipv6 V6_INPUT_FILTER`

2. Add an ACE that denies access to an IP addresses 1001::2 through 2001::2 (includes Host B):

   `switch(config-acl-ipv6)# deny any 1001::2 2001::2 log`

3. Add an ACE that allows access from all other IP addresses:

   `switch(config-acl-ipv6)# permit any any any`

4. Exit the ACL definition:

   `switch(config-acl-ipv6)# exit`

5. Enter the interface to which you will apply the ACL:

   `switch(config)# interface 1/1/1`

6. Apply the V6_INPUT_FILTER ACL to inbound (ingress) traffic:

   `switch(config-if)# apply access-list ipv6 V6_INPUT_FILTER in`

7. Show your ACL:

```
switch(config-if)# exit
switch# show access-list interface 1/1/1
```

```
Direction
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
----------------------------------------------------------------------------
Inbound
IPv6       V6_INPUT_FILTER
        10
           deny                            any
           1001::2
           2001::2
           Logging: enabled
           Hit-counts: enabled
        20
           permit                          any
           any
           any
----------------------------------------------------------------------------
```

Chapter 4
Classifier policies

Classifier policies let a network administrator define sets of rules based on network traffic addressing or other header content, and use these rules to restrict or alter the passage of traffic through the switch. Choosing the rule criteria is called Classification, and one such rule, or list, is called a policy. Classification is achieved by creating a traffic class. The types of classes (IPv4, and IPv6) are each focused on relevant frame/packet characteristics. Classes can be configured to match or ignore almost any frame or packet header field. Network traffic passing through a switch can be classified based on many different frame/packet characteristics including, but not limited to:

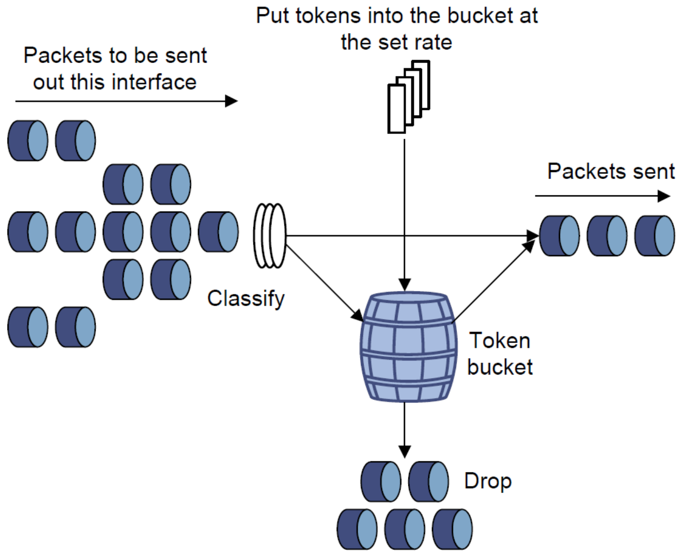
- Frame ingress VLAN ID
- Source and/or destination IPv4, or IPv6 address
- Layer 2 (EtherType) and Layer 3 (IP) protocol
- Layer 4 application ports

A policy contains one or more policy entries, which are listed according to priority by sequence number. A single policy entry contains a class and corresponding policy action. Policy action is taken on traffic matched by its corresponding class. A policy can be applied to an individual front plane port, a Link Aggregation Group (LAG) interface, or a VLAN.

See also ACL and Policy hardware resource considerations.

Traffic policing

Traffic policing supports policing of the inbound traffic. A typical application of traffic policing is to supervise the specification of traffic entering a network and limit it within a reasonable range. Another application is to "discipline" the extra traffic to prevent aggressive use of network resources by an application. For example, you can limit bandwidth for HTTP packets to less than 50% of the total. If the traffic of a session exceeds the limit, traffic policing can drop the packets or reset the IP precedence of the packets. In the following illustrated example, outbound traffic is policed:



Traffic policing is widely used in policing traffic entering the ISP networks. It can classify the policed traffic and take predefined policing actions on each packet depending on the evaluation result:

- Forwarding the packet if the evaluation result is "conforming."
- Dropping the packet if the evaluation result is "excess."
- Forwarding the packet with its precedence remarked when the evaluation result is "conforming."

Types of policy actions

The policy actions are broadly classified in the following categories:

- Remark actions
- Police actions
- Other actions

Each policy entry can have a combination of policy actions from these multiple categories, which are executed in the order of configuration.

Remark actions

This category contains the following actions:

- **Priority Code Point (PCP):** 3-bit field in layer 2 802.1Q header refers to a class of service and maps to a frame priority level.
- **IP precedence:** 3-bit field in IP header which denotes the importance or priority of the datagram.
- **IP Differentiated Services Code Point (DSCP):** 6-bit field in IP header for packet classification.
- **Local Priority:** Change the internal priority used to queue the packets for transmission. Local priority can be used to rewrite the priority of traffic classes local to the system based on the QoS mapping settings without changing the IP header or the 802.1Q header. Remark actions other than local priority only change packets as they leave the switch. The local priority action can be combined with the other remark actions to remark packets and change the internal priority to reflect the new priority.

Police actions

Traffic policing meters inbound traffic on an interface or VLAN based on the following traffic parameters:

- Committed information rate (CIR): Bandwidth limit for guaranteed traffic.
- Committed burst size (CBS): Maximum packet size permitted for bursts of data that exceed the CIR.

Based on these parameters, packets are dropped when traffic exceeds the bandwidth limit (CIR) and the burst size for guaranteed traffic (CBS).

Other actions

Other actions include Drop: Drop the packet, and Mirror: Mirror the packets to a specified mirroring session. For details, see the *Monitoring Guide*.

How policy matching works

A policy can be applied to an interface or VLAN to affect/control traffic arriving on that interface or VLAN (inbound (ingress)). A single policy entry matches on one or more characteristics of the particular traffic type and has a configured action to continue through the switch. This matching occurs by beginning with the entry with the lowest sequence number. The entry is then compared against the incoming frame to its particular match characteristics. If there is a match, the action is taken.

If there is no match, the match characteristics of the next sequence are compared to the relevant frame/packet details. If there is a match, the specified actions are taken. This process continues until a match is found; otherwise, the packet is permitted to flow through the switch unaltered. The "implicit permit" behavior of policy matching differs from the "implicit deny" behavior of ACL matching.

Limitations

The following match criteria are not supported:

Active class configuration versus user-specified configuration

The output of the `show class` command displays the active class configurations. Active class configurations are the classes that have been configured and accepted by the system.

The output of the `show class` command with the `configuration` parameter, displays the classes that have been configured by the user.

Discrepancies might occur between the active class configurations and the user-specified configurations. In the user-specified class configurations, unsupported command parameters may have been configured, or class can be modified after policy application and may have been unsuccessful due to lack of hardware resources.

To determine if a discrepancy exists between what was configured and what is active, run any variant of the `show class` command. If the active classes and configured classes are not the same, a warning message is displayed.

```
! class MY_CLASS user configuration does not match active configuration.
! run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the configured class is processing and you entered the `show class` command with parameters, the following in-progress message is displayed:

```
! class ip MY_CLASS user configuration currently being processed
! run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the configured class is processing and you entered the `show class` command without parameters, the following in-progress message is displayed:

```
% Warning: MY_CLASS user configuration currently being processed
% run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the warning message or in-progress message is displayed, additional changes may be made until the error message is no longer displayed. Or you can use the `class {all|ip <class-name>|ipv6 <class-name>|mac <class-name>} reset` command to change the user-specified configuration to match the active configuration.

The `show running-config` command also shows a warning about classes that are in progress or failed.

Example

Resetting the user-specified class configuration to the active configuration:

```
switch(config)# class all reset
```

Active policy configuration versus user-specified configuration

The output of the `show policy` command displays the active policy configurations. Active policy configurations are the policies that have been configured and accepted by the system. With applied policies, the active configuration displays the interfaces on which the policies have successfully been programmed in hardware.

The output of the `show policy` command with the `configuration` parameter, displays the policies that have been configured by the user.

Discrepancies might exist between the active policy configurations and the user-specified configurations. In the user-specified policy configurations, unsupported command parameters might have been configured, or an application of a policy might have been unsuccessful because of a lack of hardware resources.

To determine if a discrepancy exists between the configuration and what is active, run any variant of the `show policy` command. If the active policies and configured policies are not the same, a warning message is displayed in the output of the `show` command.

```
! policy MY_POLICY user configuration does not match active configuration.
! run 'policy NAME reset' to reset policy to match active configuration.
```

The switch displays an `in progress` message while it processes the configured policy:

```
! policy MY_POLICY user configuration currently being processed
! run 'policy NAME reset' to reset policy to match active configuration.
```

If the warning message or in progress message is displayed, additional changes may be made until the error message is no longer displayed. Or you can use the `policy <policy-name> reset` command to change the user-specified configuration to match the active configuration.

Example

Resetting `MY_POLICY`:

```
switch(config)# policy MY_POLICY reset
```

Considerations for when a policy is applied per interface

This section is only applicable to polices applied to physical interfaces and LAGs using the `per-interface` parameter.

The `reset` command (mentioned in the previous section) is not useful if one or more unique instances of a policy created using the `per-interface` parameter fail to update in hardware even though the parent policy does update. If this occurs, you can make additional changes to the policy and its applications to correct the discrepancy until the error messages are no longer displayed. Alternatively consider using command `checkpoint-rollback` as described in the *AOS-CX Fundamentals Guide*.

Policies using the `per-interface` parameter have slightly different warning and in-progress messages due to unique instances of the policy being created and applied to individual physical interfaces and LAGs.

For example, this is how the warning messages will appear if the unique instances of the policy for interfaces 1/1/2-1/1/3 fail to update while the unique instances of the policy for interfaces 1/1/1,1/1/4 successfully update.

```
switch(config)# show policy commands
! policy my_policy user configuration does not match active configuration on
interface 1/1/2 for ingress.
! policy my_policy user configuration does not match active configuration on
interface 1/1/3 for ingress.
policy my_policy
    10 class ip my_ip_class action drop
interface 1/1/1
    apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
interface 1/1/2
    apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
interface 1/1/3
    apply policy my_policy in per-interface
interface 1/1/4
    apply policy my_policy in per-interface


switch(config)# show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
% Warning: my_policy user configuration does not match active configuration on
interface 1/1/2 for ingress.
% Warning: my_policy user configuration does not match active configuration on
```

```
interface 1/1/3 for ingress.
           my_policy
        10
           my_ip_class ipv4
                    drop
```

This is how the in-progress messages will appear if the child policies for interfaces 1/1/2-1/1/3 are currently updating while the child policies for interfaces 1/1/1,1/1/4 have successfully updated.

```
switch(config)# show policy commands
! policy my_policy user configuration currently being processed on interface 1/1/2
for ingress.
! policy my_policy user configuration currently being processed on interface 1/1/3
for ingress.
! run 'show policy [commands]' to display active policy configuration.
policy my_policy
    10 class ip my_ip_class action drop
interface 1/1/1
    apply policy my_policy in per-interface
! policy my_policy user configuration currently being processed
! run 'show policy [commands]' to display active policy configuration.
interface 1/1/2
    apply policy my_policy in per-interface
! policy my_policy user configuration currently being processed
! run 'show policy [commands]' to display active policy configuration.
interface 1/1/3
    apply policy my_policy in per-interface
interface 1/1/4
    apply policy my_policy in per-interface

switch(config)# show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
% Warning: my_policy user configuration currently being processed on interface
1/1/2 for ingress.
% Warning: my_policy user configuration currently being processed on interface
1/1/3 for ingress.
%          run 'show policy [commands]' to display active policy configuration.
           my_policy
        10
           my_ip_class ipv4
                    drop
```

This is how the warning messages will appear if the child policies for interfaces 1/1/2-1/1/3 failed to apply or replace while the child policies for interfaces 1/1/1,1/1/4 have successfully applied or replaced.

```
switch(config)# show policy commands
policy my_policy
    10 class ip my_ip_class action drop
interface 1/1/1
    apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
! run 'policy NAME reset' to reset policy to match active configuration.
interface 1/1/2
    apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
```

```
! run 'policy NAME reset' to reset policy to match active configuration.
interface 1/1/3
    apply policy my_policy in per-interface
interface 1/1/4
    apply policy my_policy in per-interface

switch(config)# show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
           my_policy
        10
           my_ip_class ipv4
                    drop
```

Classifier policy commands

Classifier policy application

Classifier policies can be applied as follows ("Rt-In" = "Routed-In"):

Policy type Direction	IPv4+6 In	IPv4+6 Rt-In	MAC In
L2 interface (port)	Yes		Yes
L2 LAG	Yes		Yes
L3 interface (port)	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes
VLAN	Yes		Yes
Interface VLAN		Yes (PBR)	

The following match criteria are not supported. If any of these match criteria are attempted to be configured, an error message will be displayed and the action will not be completed.

```
TCP flags CWR and ECE
TCP flags and TTL (hop limit) on IP classes
Fragment on IPv6 classes in VLAN policies
VLAN ID in VLAN policies
```

apply policy (config-if, config-lag-if, config-if-vlan, config-vlan)

Context config-if, config-lag-if:

```
apply policy <POLICY-NAME> {in|out|routed-in} [per-interface]
no apply policy <POLICY-NAME> {in|out|routed-in} [per-interface]
```

Context config-vlan:

```
apply policy <POLICY-NAME> {in|out}
no apply policy <POLICY-NAME> {in|out}
```

Context config-if-vlan:

```
apply policy <POLICY-NAME> routed-in
no apply policy <POLICY-NAME> routed-in
```

Description

Applies a policy to the current physical interface port or LAG or VLAN context.

Only one direction of a policy can be applied to an interface or VLAN at a time, thus using the apply command on an interface or VLAN with an already-applied policy of the same direction will replace the currently applied policy.

> The VLAN context supports the **in** and **out** directions, which apply to both bridged and routed traffic. The Interface VLAN context only supports the **routed-in** direction which applies only to routed traffic.

The **no** form of this command removes a policy from the interface or VLAN specified by the current context.

Parameter	Description
<POLICY-NAME>	Specifies the policy to apply.
`in`	Selects the inbound (ingress) traffic direction.
`out`	Selects the outbound (egress) traffic direction.
`routed-in`	Selects routed in traffic.
`per-interface`	Specifies that unique instances of the policy be applied to each interface or LAG rather than the default of sharing the policy across all interfaces and LAGs.

Usage (applies to *config-if*, *config-lag-if* contexts)

- When **per-interface** is included, unique instances of the policy are applied to each physical interface port or LAG rather than the default of sharing the policy across all interfaces and LAGs. The unique instance of a policy has a parent-child relationship with the policy from which it was created. The **per-interface** option is useful when you want unique policers to be created for each interface or LAG rather than using shared policers. It is also useful when you want the statistics (hit counts and conform rate) to be specific to an interface or LAG rather than being aggregated. Because **per-interface** creates more hardware instances of a policy, resource consumption may increase significantly. It is recommended that you use **show resources** to monitor resource utilization as configuration is applied.

Usage (applies to config-vlan context)

- Only one policy type may be applied to a VLAN at a time. Therefore, using the **apply policy** command on a VLAN with an already-applied policy of the same type, will replace the applied policy.
- When a policy is applied to a VLAN, it will create hardware entries on all line cards regardless of whether a VLAN member exists on any specific line card.

Examples

Applying a policy to an interface (ingress):

```
switch(config)# interface 1/1/1
switch(config-if)# apply policy MY_POLICY1 in
```

Applying a policy to an interface (ingress) specifying **per-interface**:

```
switch(config)# interface 1/1/2
switch(config-if)# apply policy MY_POLICY1 in per-interface
```

Applying a policy to an interface (egress):

```
switch(config)# interface 1/1/2
switch(config-if)# apply policy MY_POLICY2 out
```

Applying a policy to an interface (egress) specifying `per-interface`:

```
switch(config)# interface 1/1/2
switch(config-if)# apply policy MY_POLICY2 out per-interface
```

Applying a policy to an interface range (ingress):

```
switch(config)# interface 1/1/3-1/1/6
switch(config-if-<1/1/2-1/1/5>)# apply policy MY_POLICY3 in
```

Applying a policy to an interface range (ingress) specifying **per-interface**:

```
switch(config)# interface 1/1/7-1/1/9
switch(config-if-<1/1/2-1/1/5>)# apply policy MY_POLICY4 in per-interface
```

Removing a policy from an interface (ingress):

```
switch(config)# interface 1/1/1
switch(config-if)# no apply policy MY_POLICY1 in
```

Removing a policy from an interface range (ingress):

```
switch(config)# interface 1/1/3-1/1/6
switch(config-if-<1/1/3-1/1/6>)# no apply policy MY_POLICY3 in
```

Applying a policy to a LAG (ingress):

```
switch(config)# interface lag 100
switch(config-lag-if)# apply policy MY_POLICY5 in
```

Applying a policy to a LAG (ingress) specifying **per-interface**:

```
switch(config)# interface lag 200
switch(config-lag-if)# apply policy MY_POLICY5 in per-interface
```

Removing a policy from a LAG (ingress):

```
switch(config)# interface lag 100
switch(config-lag-if)# no apply policy MY_POLICY5 in
```

Applying a policy to a VLAN (ingress):

```
switch(config)# vlan 1
switch(config-vlan)# apply policy MY_POLICY6 in
```

Applying a policy to a VLAN (egress):

```
switch(config)# vlan 1
switch(config-vlan)# no apply policy my_policy out
```

Applying a policy to multiple VLANs (ingress):

```
switch(config)# vlan 10,20
switch(config-vlan-<10,20>)# apply policy MY_POLICY7 in
```

Applying a policy to an interface VLAN routed (ingress):

```
switch(config)# vlan 1
switch(config-if-vlan)# apply policy MY_POLICY8 routed-in
```

Applying a policy to an interface VLAN range routed (ingress):

```
switch(config)# vlan 2-5
switch(config-if-vlan-<2-5>)# apply policy MY_POLICY8 routed-in
```

Removing a policy from a VLAN (ingress):

```
switch(config)# vlan 1
switch(config-vlan)# no apply policy MY_POLICY6 in
```

Removing a policy from multiple VLANs (ingress):

```
switch(config)# vlan 10,20
switch(config-vlan-<10,20>)# no apply policy MY_POLICY7 in
```

Removing a policy from an interface VLAN routed (ingress):

```
switch(config)# vlan 1
switch(config-if-vlan)# no apply policy MY_POLICY8 routed-in
```

Removing a policy from an interface VLAN range routed (ingress):

```
switch(config)# vlan 2-5
switch(config-if-vlan-<2-5>)# no apply policy MY_POLICY8 routed-in
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority

All platforms	`config-if`
	`config-lag-if`
	`config-vlan`
	`config-if-vlan`
		Administrators or local user group members with execution rights for this command.

class copy

`class {ip|ipv6|mac} <CLASS-NAME> copy <DESTINATION-CLASS>`

Description

Copies a class to a new destination class or overwrites an existing class. Copying a class copies all entries as well.

Parameter	Description
{ip|ipv6|mac**}** *<CLASS-NAME>*	Specifies the type and name of the class to be copied.
<DESTINATION-CLASS>	Specifies the name of the destination class.

Examples

Copying an IPv4 class. Copying a class with entries copies all its entries as well:

```
switch(config)# class ip MY_IP_CLASS copy MY_IP_CLASS2
switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_CLASS
       11  ignore                          udp
```

```
          any
          any
      21  match                          tcp
          192.168.0.1
          192.168.0.2
-------------------------------------------------------------------------------
IPv4      MY_IP_CLASS2
      11  ignore                         udp
          any
          any
      21  match                          tcp
          192.168.0.1
          192.168.0.2
```

Copying an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS copy MY_IPV6_CLASS2
switch(config)# do show class
Type      Name
  Sequence Comment
          Action                         L3 Protocol
          Source IP Address              Source L4 Port(s)
          Destination IP Address         Destination L4 Port(s)
          Additional Parameters
-------------------------------------------------------------------------------
IPv6      MY_IPV6_CLASS
        2 ignore                         udp
          any
          any
-------------------------------------------------------------------------------
IPv6      MY_IPV6_CLASS2
        2 ignore                         udp
          any
          any
```

Copying a MAC class:

```
switch(config)# class mac MY_MAC_CLASS copy MY_MAC_CLASS2
switch(config)# do show class
Type      Name
  Sequence Comment
          Action                         EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-------------------------------------------------------------------------------
MAC       MY_MAC_CLASS
        2 ignore                         arp
          any
          any
-------------------------------------------------------------------------------
MAC       MY_MAC_CLASS2
        2 ignore                         arp
          any
          any
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

class ip

Syntax to create an IPv4 class and enter its context. Plus syntax to remove a class:

```
class ip <CLASS-NAME>
no class ip <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols **ah**, **gre**, **esp**, **igmp**, **ospf**, **pim** (**ip** is available as an alias for **any**):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols **sctp**, **tcp**, **udp**:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[cwr][ece] [urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>]  [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol **icmp**:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{icmp}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv4 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, the class can classify traffic based on IPv4 header information.

The **no** form of the command can be used to delete either an IPv4 traffic class (use **no** with the class command) or an individual IPv4 traffic class entry (use **no** with the sequence number).

Parameter	Description
`ip`	Specifies create or modify an IPv4 class.
`<CLASS-NAME>`	Specifies the name of this class.
`<SEQUENCE-NUMBER>`	Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.
`{match|ignore}`	Creates a rule to match or ignore specified packets.
`<IP-PROTOCOL-NUM>`	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
`{any|<SRC-IP-ADDRESS> [/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}`	Specifies the source IPv4 address. ■ **any** - specifies any source IPv4 address. ■ **<SRC-IP-ADDRESS>** - specifies the source IPv4 host address. ◦ **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ **<SUBNET-MASK>** - specifies the address bits to mask (dotted decimal notation).
`{any|<DST-IP-ADDRESS> [/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}`	Specifies the destination IPv4 address. ■ **any** - specifies any destination IPv4 address. ■ **<DST-IP-ADDRESS>** - specifies the destination IPv4 host address. ◦ **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ **<SUBNET-MASK>** - specifies the address bits to mask (dotted decimal notation).
`[{eq|gt|lt} <PORT>|range <MIN-PORT><MAX-PORT>]`	Specifies the port or port range. Port numbers are in the range of 0 to 65535. ■ **eq <PORT>** - specifies the Layer 4 port. ■ **gt <PORT>** - specifies any Layer 4 port greater than the indicated port. ■ **lt <PORT>** - specifies any Layer 4 port less than the indicated port. ■ **range *<MIN-PORT> <MAX-PORT>*** - specifies the Layer 4 port range.

Parameter	Description
`cwr`	Specifies matching on the TCP Flag CWR : Congestion Window Reduced
`ece`	Specifies matching on the TCP Flag ECE : Explicit Congestion Notification [ECN]- Echo
`urg`	Specifies matching on the TCP Flag: Urgent.
`ack`	Specifies matching on the TCP Flag: Acknowledgment.
`psh`	Specifies matching on the TCP Flag: Push buffered data to receiving application.
`rst`	Specifies matching on the TCP Flag: Reset the connection.
`syn`	Specifies matching on the TCP Flag: Synchronize sequence numbers.
`fin`	Specifies matching on the TCP Flag: Finish connection.
`established`	Specifies matching on the TCP Flag: Established connection.
`dscp` *`<DSCP-SPECIFIER>`*	Specifies the Differentiated Services Code Point (DSCP), either a numeric **`<DSCP-VALUE>`** (0 to 63) or one of these keywords: ■ **AF11** - DSCP 10 (Assured Forwarding Class 1, low drop probability) ■ **AF12** - DSCP 12 (Assured Forwarding Class 1, medium drop probability) ■ **AF13** - DSCP 14 (Assured Forwarding Class 1, high drop probability) ■ **AF21** - DSCP 18 (Assured Forwarding Class 2, low drop probability) ■ **AF22** - DSCP 20 (Assured Forwarding Class 2, medium drop probability) ■ **AF23** - DSCP 22 (Assured Forwarding Class 2, high drop probability) ■ **AF31** - DSCP 26 (Assured Forwarding Class 3, low drop probability) ■ **AF32** - DSCP 28 (Assured Forwarding Class 3, medium drop probability) ■ **AF33** - DSCP 30 (Assured Forwarding Class 3, high drop probability) ■ **AF41** - DSCP 34 (Assured Forwarding Class 4, low drop probability) ■ **AF42** - DSCP 36 (Assured Forwarding Class 4, medium drop probability) ■ **AF43** - DSCP 38 (Assured Forwarding Class

Parameter	Description
	4, high drop probability) ■ **CS0** - DSCP 0 (Class Selector 0: Default) ■ **CS1** - DSCP 8 (Class Selector 1: Scavenger) ■ **CS2** - DSCP 16 (Class Selector 2: OAM) ■ **CS3** - DSCP 24 (Class Selector 3: Signaling) ■ **CS4** - DSCP 32 (Class Selector 4: Realtime) ■ **CS5** - DSCP 40 (Class Selector 5: Broadcast video) ■ **CS6** - DSCP 48 (Class Selector 6: Network control) ■ **CS7** - DSCP 56 (Class Selector 7) ■ **EF** - DSCP 46 (Expedited Forwarding)
`ecn` *<ECN-VALUE>*	Specifies an Explicit Congestion Notification value. Range: 0 to 3.
`ip-precedence` *<IP-PRECEDENCE-VALUE>*	Specifies an IP precedence value. Range: 0 to 7.
`tos` *<TOS-VALUE>*	Specifies the Type of Service value. Range: 0 to 31.
`fragment`	Specifies a fragment packet.
`vlan` *<VLAN-ID>*	Specifies VLAN tag to match on. 802.1Q VLAN ID. **NOTE:** This parameter cannot be used in any class that will be applied to a VLAN.
`ttl` *<TTL-VALUE>*	Specifies a time-to-live (hop limit) value. Range: 0 to 255.
`count`	Keeps the hit counts of the number of packets matching this class entry.
[*<SEQUENCE-NUMBER>*] `comment` *<TEXT-STRING>*	Adds a comment to a class entry. The **no** form removes only the comment from the class entry.

Usage

- Entering an existing **<CLASS-NAME>** value will cause the existing class to be modified, with any new **<SEQUENCE-NUMBER>** value creating an additional class entry, and any existing **<SEQUENCE-NUMBER>** value replacing the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry will be appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the **<IP-PROTOCOL-NUM>** parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

The following match criteria are not supported on HPE Aruba Networking 8400 Switch series:

- TCP flags **CWR** and **ECE**
- **TTL** (hop limit)
- A VLAN ID in VLAN policies

Examples

Creating an IPv4 class with three entries:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match icmp any any10 match icmp any any
switch(config-class-ip)# 20 ignore udp any any
switch(config-class-ip)# 30 match tcp 192.168.0.1 192.168.0.2
switch(config-class-ip)# exit

switch(config)# do show class
Type        Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4        MY_IP_CLASS
        10 match                           icmp
           any
           any
        20 ignore                          udp
           any
           any
        30 match                           tcp
           192.168.0.1
           192.168.0.2
```

Adding a comment to an existing IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 30 comment myipClass
switch(config-class-ip)# exit

switch(config)# do show class
Type        Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4        MY_IP_CLASS
        10 match                           icmp
           any
           any
        20 ignore                          udp
           any
           any
        30 myipClass
           match                           tcp
           192.168.0.1
           192.168.0.2
```

Removing a comment from an existing IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 30 comment
switch(config-class-ip)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_CLASS
        10 match                           icmp
           any
           any
        20 ignore                          udp
           any
           any
        30 match                           tcp
           192.168.0.1
           192.168.0.2

Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_CLASS
        10 match                           icmp
           any
           any
        20 ignore                          udp
           any
           any
        30 match                           tcp
           192.168.0.1
           192.168.0.2
```

Replacing an IPv4 class entry in an existing class:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match igmp any any
switch(config-class-ip)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv4       MY_IP_CLASS
        10 match                           igmp
           any
           any
        20 ignore                          udp
           any
```

```
         any
      30 match                          tcp
         192.168.0.1
         192.168.0.2
```

Removing an IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 10
switch(config-class-ip)# exit

switch(config)# do show class
Type        Name
  Sequence Comment
            Action                          L3 Protocol
            Source IP Address               Source L4 Port(s)
            Destination IP Address          Destination L4 Port(s)
            Additional Parameters
-------------------------------------------------------------------------------
IPv4        MY_IP_CLASS
       20   ignore                          udp
            any
            any
       30   match                           tcp
            192.168.0.1
            192.168.0.2
```

Removing an IPv4 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.

The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ip MY_IP_CLASS

switch(config)# do show class
No Class found.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` **The class ip <CLASS-NAME> command takes you into the config-class-ipconfig-**	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
	class-ip context where you enter the class entries.	

class ipv6

Syntax to create an IPv6 class and enter its context. Plus syntax to remove a class:

```
class ipv6 <CLASS-NAME>
no class ipv6 <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols `ah`, `gre`, `esp`, `igmp`, `ospf`, `pim` (`ipv6` is available as an alias for `any`):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ipv6|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols `sctp`, `tcp`, `udp`:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>]  [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>]  [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol `icmpv6`:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ecn <ECN-VALUE>]  [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [ttl <TTL-VALUE>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv6 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on IPv6 header information.

The **no** form of the command deletes either an IPv6 traffic class (use **no** with the class command) or an individual IPv6 traffic class entry (use **no** with the sequence number).

Parameter	Description
ipv6	Specifies create or modify an IPv6 class.
<CLASS-NAME>	Specifies the name of this class.
<SEQUENCE-NUMBER>	Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.
{match|ignore}	Creates a rule to match or ignore specified packets.
<IP-PROTOCOL-NUM>	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
{any|*<SRC-IP-ADDRESS>*[/ { *<PREFIX-LENGTH>*|*<SUBNET-MASK>*}]}	Specifies the source IPv6 address. ■ **any** - specifies any source IPv6 address. ■ **<SRC-IP-ADDRESS>** - specifies the source IPv4 host address. ◦ **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ **<SUBNET-MASK>** - specifies the address bits to mask (dotted decimal notation).
{any|*<DST-IP-ADDRESS>*[/ { *<PREFIX-LENGTH>*|*<SUBNET-MASK>*}]}	Specifies the destination IPv4 address. ■ **any** - specifies any destination IPv6 address. ■ **<DST-IP-ADDRESS>** - specifies the destination IPv6 host address. ◦ **<PREFIX-LENGTH>** - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ **<SUBNET-MASK>** - specifies the address bits to mask (dotted decimal notation).
[{eq|gt|lt} *<PORT>*|range *<MIN-PORT><MAX-PORT>*]	Specifies the port or port range. Port numbers are in the range of 0 to 65535. ■ **eq <PORT>** - specifies the Layer 4 port. ■ **gt <PORT>** - specifies any Layer 4 port greater than the indicated port. ■ **lt <PORT>** - specifies any Layer 4 port less than the indicated port. ■ **range <MIN-PORT> <MAX-PORT>** - specifies the Layer 4 port range.
cwr	Specifies matching on the TCP Flag CWR : Congestion Window Reduced
ece	Specifies matching on the TCP Flag ECE : Explicit Congestion Notification [ECN]- Echo
urg	Specifies matching on the TCP Flag: Urgent.

Parameter	Description
`ack`	Specifies matching on the TCP Flag: Acknowledgment.
`psh`	Specifies matching on the TCP Flag: Push buffered data to receiving application.
`rst`	Specifies matching on the TCP Flag: Reset the connection.
`syn`	Specifies matching on the TCP Flag: Synchronize sequence numbers.
`fin`	Specifies matching on the TCP Flag: Finish connection.
`established`	Specifies matching on the TCP Flag: Established connection.
`dscp` *<DSCP-SPECIFIER>*	Specifies the Differentiated Services Code Point (DSCP), either a numeric **<DSCP-VALUE>** (0 to 63) or one of these keywords: ■ **AF11** - DSCP 10 (Assured Forwarding Class 1, low drop probability) ■ **AF12** - DSCP 12 (Assured Forwarding Class 1, medium drop probability) ■ **AF13** - DSCP 14 (Assured Forwarding Class 1, high drop probability) ■ **AF21** - DSCP 18 (Assured Forwarding Class 2, low drop probability) ■ **AF22** - DSCP 20 (Assured Forwarding Class 2, medium drop probability) ■ **AF23** - DSCP 22 (Assured Forwarding Class 2, high drop probability) ■ **AF31** - DSCP 26 (Assured Forwarding Class 3, low drop probability) ■ **AF32** - DSCP 28 (Assured Forwarding Class 3, medium drop probability) ■ **AF33** - DSCP 30 (Assured Forwarding Class 3, high drop probability) ■ **AF41** - DSCP 34 (Assured Forwarding Class 4, low drop probability) ■ **AF42** - DSCP 36 (Assured Forwarding Class 4, medium drop probability) ■ **AF43** - DSCP 38 (Assured Forwarding Class 4, high drop probability) ■ **CS0** - DSCP 0 (Class Selector 0: Default) ■ **CS1** - DSCP 8 (Class Selector 1: Scavenger) ■ **CS2** - DSCP 16 (Class Selector 2: OAM) ■ **CS3** - DSCP 24 (Class Selector 3: Signaling) ■ **CS4** - DSCP 32 (Class Selector 4: Real time) ■ **CS5** - DSCP 40 (Class Selector 5: Broadcast video) ■ **CS6** - DSCP 48 (Class Selector 6: Network control)

Parameter	Description
	■ **CS7** - DSCP 56 (Class Selector 7) ■ **EF** - DSCP 46 (Expedited Forwarding)
`ecn <ECN-VALUE>`	Specifies an Explicit Congestion Notification value. Range: 0 to 3.
`ip-precedence <IP-PRECEDENCE-VALUE>`	Specifies an IP precedence value. Range: 0 to 7.
`tos <TOS-VALUE>`	Specifies the Type of Service value. Range: 0 to 31.
`fragment`	Specifies a fragment packet.
`vlan <VLAN-ID>`	Specifies VLAN tag to match on. 802.1Q VLAN ID. **NOTE:** This parameter cannot be used in any class that will be applied to a VLAN.
`ttl <TTL-VALUE>`	Specifies a time-to-live (hop limit) value. Range: 0 to 255.
`count`	Keeps the hit counts of the number of packets matching this class entry.
`[<SEQUENCE-NUMBER>] comment <TEXT-STRING>`	Adds a comment to a class entry. The **no** form removes only the comment from the class entry.

Usage

- If you enter an existing **<CLASS-NAME>** value, the existing class is modified with any new **<SEQUENCE-NUMBER>** value. This action creates an additional class entry. Any existing **<SEQUENCE-NUMBER>** value replaces the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry is appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the **<IP-PROTOCOL-NUM>** parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

The following match criteria are not supported on HPE Aruba Networking 8400 Switch series:

- TCP flags **CWR** and **ECE**
- **TTL** (hop limit) on IPv6 classes '
- TCP flags on IPv6 classes
- **Fragment** on IPv6 classes in VLAN policies
- A VLAN ID in VLAN policies

Examples

Creating an IPv6 class with two entries:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
```

```
switch(config-class-ipv6)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
        10 match                           icmpv6
           any
           any
        20 ignore                          udp
           any
           any
```

Adding a comment to an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# 20 comment myipv6class
switch(config-class-ipv6)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
        10 match                           icmpv6
           any
           any
        20 myipv6class
           ignore                          udp
           any
           any
```

Removing a comment from an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 20 comment
switch(config-class-ipv6)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          L3 Protocol
           Source IP Address               Source L4 Port(s)
           Destination IP Address          Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
```

```
        10 match                            icmpv6
           any
           any
        20 ignore                           udp
           any
           any

Type       Name
  Sequence Comment
           Action                           L3 Protocol
           Source IP Address                Source L4 Port(s)
           Destination IP Address           Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
        10 match                            icmpv6
           any
           any
        20 ignore                           udp
           any
           any
```

Replacing an IPv6 class entry in an existing IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match any any 1020::
switch(config-class-ipv6)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                           L3 Protocol
           Source IP Address                Source L4 Port(s)
           Destination IP Address           Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
        10 match                            any
           any
           1020::
        20 ignore                           udp
           any
           any
```

Removing an IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 10
switch(config-class-ipv6)# exit

switch(config)# do show class
Type       Name
  Sequence Comment
           Action                           L3 Protocol
           Source IP Address                Source L4 Port(s)
           Destination IP Address           Destination L4 Port(s)
           Additional Parameters
-------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
```

```
        20 ignore                           udp
           any
           any
```

Removing an IPv6 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.

The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ipv6 MY_IPV6_CLASS

switch(config)# do show class
No Class found.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **class ipv6 <CLASS-NAME>** command takes you into the **config-class-ipv6** command context where you enter the class entries.	Administrators or local user group members with execution rights for this command.

class mac

```
class mac <CLASS-NAME>

    [<SEQUENCE-NUMBER>]
    {match|ignore}
    {any}
    {any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>}]}
    {any|aarp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|ipx-arpa|ipx-non-arpa|is-is|
          lldp|mpls-multicast|mpls-unicast|q-in-q|rbridge|trill|wake-on-lan|
          <NUMERIC-ETHERTYPE>}
    [pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count]

    [<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

Description

Creates or modifies a MAC traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on MAC header information.

The **no** form of the command can be used to delete either a MAC traffic class (use **no** with the class command) or an individual MAC traffic class entry (use **no** with the sequence number).

Parameter	Description
`mac`	Specifies create or modify a MAC class.
<CLASS-NAME>	Specifies the name of this class.
<SEQUENCE-NUMBER>	Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.
`{match|ignore}`	Creates a rule to match or ignore specified packets.
`comment`	Stores the remaining entered text as a class comment.
`{any|`*<SRC-MAC-ADDRESS>* `[/`*<ETHERNET-MASK>*`}]}`	Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword **any**. You can optionally include the following: **<ETHERNET-MASK>** - The address bits to mask (xxxx.xxxx.xxxx).
`{any|`*<DST-MAC-ADDRESS>* `[/`*<ETHERNET-MASK>*`}]}`	Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword **any**. You can optionally include the following: **<ETHERNET-MASK>** - The address bits to mask (xxxx.xxxx.xxxx).
`Protocol`	Select an ethertype protocol from the following (enter one only): ■ **any** - Any ethertype protocol ■ **<NUMERIC-ETHERTYPE>** - Enter an EtherType protocol number. Range: 0x600-0xffff. ■ Or enter an EtherType protocol name from the following list: ◦ **aarp** ◦ **appletalk** ◦ **arp** ◦ **fcoe** ◦ **fcoe-init** ◦ **ip** ◦ **ipv6** ◦ **ipx-arpa** ◦ **ipx-non-arpa** ◦ **is-is** ◦ **lldp** ◦ **mpls-multicast** ◦ **mpls-unicast** ◦ **q-in-q** ◦ **rbridge** ◦ **trill** ◦ **wake-on-lan**
`pcp` *<PCP-VALUE>*	Specifies matching on QoS Priority Code Point. Range: 0-7.
`vlan` *<VLAN-ID>*	Specifies matching on a VLAN ID. Enter a VLAN ID or the VLAN name, if configured.

Parameter	Description
	NOTE: This parameter cannot be used in any class that will be applied to a VLAN.
`count`	Keeps the hit counts of the number of packets matching this class entry.

Examples

Creating a MAC class:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# match any any lldp
switch(config-class-mac)# ignore any any arp
switch(config-class-mac)# exit
switch(config)# do show class
Type        Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC         MY_MAC_CLASS
        10 match                           lldp
           any
           any
        20 ignore                          arp
           any
           any
```

Adding a comment to an existing MAC class entry:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# 10 comment MY_CLASS_ENTRY10 comment MY_CLASS_ENTRY
switch(config-class-mac)# exit
switch(config)# do show class
Type        Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC         MY_MAC_CLASS
        10 MY_CLASS_ENTRY
           match                           lldp
           any
           any
        20 ignore                          arp
           any
           any
```

Removing a comment from an existing MAC class entry:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# no 10 comment MY_CLASS_ENTRY
switch(config-class-mac)# exit
switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_CLASS
        10 match                           lldp
           any
           any
        20 ignore                          arp
           any
           any
```

Replacing a MAC class entry in an existing MAC class:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# 10 match any any any
switch(config-class-mac)# exit
switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_CLASS
        10 match                           any
           any
           any
        20 ignore                          arp
           any
           any
```

Removing a MAC class entry:

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# no 1
switch(config-class-mac)# exit
switch(config)# do show class
Type       Name
  Sequence Comment
           Action                          EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
-------------------------------------------------------------------------------
MAC        MY_MAC_CLASS
         2 ignore                          arp
           any
           any
```

Removing a MAC class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.

The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class mac MY_MAC_CLASS
switch(config)# do show class
No Class found.
```

Command History

Release	Modification

Command Information

Platforms	Command context	Authority
8400	`config` The **class mac <CLASS-NAME>** command takes you into the **config-class-mac** context where you enter the class entries.	Administrators or local user group members with execution rights for this command.

class resequence

`class {ip|ipv6|mac} <CLASS-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>`

Description

Resequence numering in an IPv4, or IPv6, or MAC class.

Parameter	Description
`{ip|ipv6|mac} <CLASS-NAME>`	Specifies the class where you want to resequence class entries.
`<STARTING-SEQUENCE-NUMBER>`	Specifies the sequence number to start resequencing from.
`<INCREMENT>`	Specifies how much to increment the sequence numbers by.

Examples

Resequencing an IPv4 class:

```
switch(config)# class ip MY_IP_CLASS resequence 1 10
switch(config)# do show class
Type        Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
```

```
--------------------------------------------------------------------------------
IPv4       MY_IP_CLASS
        1  match                          igmp
           any
           any
       11  ignore                         udp
           any
           any
       21  match                          tcp
           192.168.0.1
           192.168.0.2
```

Resequencing an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS resequence 1 1
switch(config-class-ipv6)# exit
switch(config)# do show class
Type       Name
  Sequence Comment
           Action                         L3 Protocol
           Source IP Address              Source L4 Port(s)
           Destination IP Address         Destination L4 Port(s)
           Additional Parameters
--------------------------------------------------------------------------------
IPv6       MY_IPV6_CLASS
         1 match                          any
           any
           1020::
         2 ignore                         udp
           any
           any
```

Resequencing a MAC class:

```
switch(config)# class mac MY_MAC_CLASS resequence 1 1
switch(config)# do show class
Type       Name
  Sequence Comment
           Action                         EtherType
           Source MAC Address
           Destination MAC Address
           Additional Parameters
--------------------------------------------------------------------------------
MAC        MY_MAC_CLASS
         1 match                          any
           any
           any
         2 ignore                         arp
           any
           any
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

class reset

`class { all | ip <CLASS-NAME> | ipv6 <CLASS-NAME> |mac <CLASS-NAME> } reset`

Description

Changes the user-specified class configuration to match the active class configuration. Use this command when there is a discrepancy between what the user configured and what is active and accepted by the system.

Parameter	Description
`{ all	ip <CLASS-NAME>| ipv6 <CLASS-NAME> |mac <CLASS-NAME> }`	Specifies either **all** classes be reset or specifies the type (**ip** for IPv4, **ipv6** for IPv6 or **mac** for MAC ACL) and name of the class to be reset.

Examples

Resetting the user-specified configuration to the active configuration:

```
switch(config)# class all reset
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

clear policy hitcounts

`clear policy hitcounts { all | [<POLICY-NAME>] [[interface <IF-NAME> [in|routed-in]] | [vlan <VLAN-ID> [in]]] | global }`

Description

Clears the policy hit count statistics.

Parameter	Description
`all`	Selects all policies.
`<POLICY-NAME>`	Specifies the policy name.
`interface <IF-NAME>`	Specifies the interface name.
`vlan <VLAN-ID>`	Specifies the VLAN.
`in`	Specifies the inbound (ingress) traffic direction.
`routed-in`	Selects the routed in traffic direction. Not applicable to a policy applied to a VLAN.
`global`	Selects the globally applied policy.

Examples

Clearing policy hit counts and then showing the policy hit counts (statistics):

```
switch# clear policy hitcounts my_policy int 1/1/1 in
switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
           Hit Count  Configuration
10 class ipv6 my_class1 action dscp af21 action drop
                   0  10 match any any any count
* policy statistics are shared among each context type (interface, VLAN).
  For routed ingress, they are only shared within the same VRF.
  Use 'policy NAME copy' to create a new policy for separate statistics.
```

Clearing the globally applied policy hit counts and then showing the global policy hit counts (statistics):

```
switch# clear policy hitcounts global
switch# show policy hitcounts global
Statistics for Policy global1:
Global Policy:
           Hit Count  Configuration
10 class ipv6 my_class1 action mirror
                   0  10 match any any any count
* policy statistics are shared among each context type (interface, VLAN).
  For routed ingress, they are only shared within the same VRF.
  Use 'policy NAME copy' to create a new policy for separate statistics.
```

Clearing hit counts for policy **MY_IPv6_Policy** applied to VLAN 10 (ingress):

```
switch# clear policy hitcounts My_IPv6_Policy vlan 10 in
```

Clearing hit counts for all policies:

```
switch# clear policy hitcounts all
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

policy

```
policy <POLICY-NAME>

    [<SEQUENCE-NUMBER>]
    class {ip|ipv6|mac} <CLASS-NAME>
          action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
          [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]

    [<SEQUENCE-NUMBER>]
    comment ...
```

Description

Creates or modifies classifier policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6/MAC class and zero or more policy actions associated with it.

A policy must be applied using the **apply** command.

The **no** form of the command can be used to delete either a policy (use **no** with the policy command) or an individual policy entry (use **no** with the sequence number).

Parameter	Description
<POLICY-NAME>	Specifies the name of the policy.
<SEQUENCE-NUMBER>	Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.
`comment`	Stores the remaining entered text as a policy entry comment.
`class {ip|ipv6|mac}` *<CLASS-NAME>*	Specifies a type of class, **ip** for IPv4, **ipv6** for IPv6 and **mac** for a MAC policy. And specifies a class name.
<REMARK-ACTIONS>	Remark actions can be any of the following options: {pcp **<PRIORITY>** | **ip-precedence <IP-PRECEDENCE_VALUE> | dscp <*DSCP-VALUE*> | local-priority <*LOCAL-PRIORITY-VALUE*>**} where:
`pcp` *<PCP-VALUE>*	Specifies the Priority Code Point (PCP) value. Range: 0 to 7.
`ip-precedence` *<IP-PRECEDENCE-VALUE>*	Specifies the numeric IP precedence value. Range: 0 to 7.

Parameter	Description
dscp *<DSCP-VALUE>*	Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows: ■ **AF11** - DSCP 10 (Assured Forwarding Class 1, low drop probability) ■ ■ **AF12** - DSCP 12 (Assured Forwarding Class 1, medium drop probability) ■ **AF13** - DSCP 14 (Assured Forwarding Class 1, high drop probability) ■ **AF21** - DSCP 18 (Assured Forwarding Class 2, low drop probability) ■ **AF22** - DSCP 20 (Assured Forwarding Class 2, medium drop probability) ■ **AF23** - DSCP 22 (Assured Forwarding Class 2, high drop probability) ■ **AF31** - DSCP 26 (Assured Forwarding Class 3, low drop probability) ■ **AF32** - DSCP 28 (Assured Forwarding Class 3, medium drop probability) ■ **AF33** - DSCP 30 (Assured Forwarding Class 3, high drop probability) ■ **AF41** - DSCP 34 (Assured Forwarding Class 4, low drop probability) ■ **AF42** - DSCP 36 (Assured Forwarding Class 4, medium drop probability) ■ **AF43** - DSCP 38 (Assured Forwarding Class 4, high drop probability) ■ **CS0** - DSCP 0 (Class Selector 0: Default) ■ **CS1** - DSCP 8 (Class Selector 1: Scavenger) ■ **CS2** - DSCP 16 (Class Selector 2: OAM) ■ **CS3** - DSCP 24 (Class Selector 3: Signaling) ■ **CS4** - DSCP 32 (Class Selector 4: Real time) ■ **CS5** - DSCP 40 (Class Selector 5: Broadcast video) ■ **CS6** - DSCP 48 (Class Selector 6: Network control) ■ **CS7** - DSCP 56 (Class Selector 7) ■ **EF** - DSCP 46 (Expedited Forwarding)
local-priority *<LOCAL-PRIORITY-VALUE>*	Specifies a local priority value. Range: 0 to 7.
<POLICE-ACTIONS>	Police actions can be the following **{cir *<RATE-BPS>*cbs <BYTES> exceed}** where:
cir kbps *<RATE-KBPS>*	Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.
cbs *<BYTES>*	Specifies a Committed Burst Size value in bytes. Range: 1 to 4294967295.
<OTHER-ACTIONS>	Other actions can be the following:
drop	Specifies drop traffic.

Usage

- An applied policy will process a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.
- Entering an existing **<POLICY-NAME>** value will cause the existing policy to be modified, with any new **<SEQUENCE-NUMBER>** value creating an additional policy entry, and any existing **<SEQUENCE-NUMBER>** value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.
- On an 8400 switch, IP remarking (**ip-precedence**, **dscp**) will only be performed if the packet is routed.

Examples

Creating a policy with several entries:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ipv6 MY_CLASS1 action dscp af21 action drop
switch(config-policy)# 20 class ip MY_CLASS3 action mirror 1
switch(config-policy)# exit
switch(config)# show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
           MY_POLICY
        10
           MY_CLASS1 ipv6
                    drop
                    dscp AF21

        20
           MY_CLASS3 ipv4
                    mirror 1
```

Adding a comment to an existing policy entry:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 20 comment MY_TEST_POLICY
switch(config-policy)# exit
switch(config)# show policy
  Name       Sequence Comment           Class       Type   action
-------------------------------------------------------------------------------
  MY_POLICY    10                       MY_CLASS1   ipv6   drop dscp AF21
               20     MY_TEST_POLICY    MY_CLASS3   ipv4   mirror 1
```

Removing a comment from an existing policy entry:

```
switch(config)# policy MY_POLICY
switch(config-policy)# no 20 comment
switch(config-policy)# exit
switch(config)# show policy
Name         Sequence  Comment  Class     Type    action
-------------------------------------------------------------------------------
MY_POLICY    10                 MY_CLASS1 ipv6    drop dscp AF21
```

```
           20                 MY_CLASS3 ipv4    mirror 1
```

Adding/Replacing a policy entry in an existing policy:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ip MY_CLASS3 action drop action dscp af21
switch(config-policy)# exit
switch(config)# show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
           MY_POLICY
        10
           MY_CLASS3 ipv4
                    drop
                    dscp AF21

        20
           MY_CLASS3 ipv4
                    mirror 1
```

Removing a policy entry:

```
switch(config)# policy MY_POLICY
switch(config-policy)# no 10
switch(config-policy)# exit
switch(config)# show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
           MY_POLICY
        20
           MY_CLASS3 ipv4
                    mirror 1
```

Removing a policy:

```
switch(config)# no policy MY_POLICY
switch(config)# show policy
Name        Sequence    Comment    Class      Type     action
-------------------------------------------------------------------------------
MY_POLICY   22                     MY_CLASS3   ipv4   mirror 1
```

The policer **exceed** DSCP action cannot be combined with other actions in the same policy entry, but other entries in the policy may use other actions.

For example, this configuration is valid:

```
switch(config)# policy my_policy
switch(config-policy)# 10 class ip my_class action cir kbps 1000 cbs 15625 exceed
dscp EF
```

But this is not because it adds a secondary action within the same policy entry:

```
6300(config-policy)# 10 class ip my_class action cir kbps 1000 cbs 15625 exceed
dscp EF action mirror 1
```

Invalid input: action

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config` The **policy** command takes you into the **config-policy** context where you enter the policy entries.	Administrators or local user group members with execution rights for this command.

policy copy

`policy <POLICY-NAME> copy <DESTINATION-POLICY>`

Description

Copies a policy to a new destination policy or overwrites an existing policy. Copying a policy copies all its entries as well.

Parameter	Description
<POLICY-NAME>	Specifies the policy to be copied.
<DESTINATION-POLICY>	Specifies the name of the destination policy.

Examples

Copying a policy:

```
switch(config)# policy MY_POLICY copy MY_POLICY2
switch(config)# do show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
           MY_POLICY
         2
           my_class3 ipv4
```

```
                    mirror 1
-------------------------------------------------------------------------------
           MY_POLICY2
         2
           my_class3 ipv4
                    mirror 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

policy resequence

`policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>`

Description

Resequences numbering in a policy.

Parameter	Description
<POLICY-NAME>	Specifies the policy where you want to resequence policy entries.
<STARTING-SEQ-NUM>	Specifies the sequence number to start resequencing from.
<INCREMENT>	Specifies how much to increment the sequence numbers by.

Examples

Resequencing a policy:

```
switch(config)# policy MY_POLICY resequence 1 1
switch(config)# do show policy
           Name
  Sequence Comment
           Class Type
                    action
-------------------------------------------------------------------------------
           MY_POLICY
         1
           MY_CLASS3 ipv4
                    drop
                    dscp AF21

         2
           MY_CLASS3 ipv4
```

```
                      mirror 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

policy reset

`policy <POLICY-NAME> reset`

Description

Changes the user-specified policy configuration to match the active policy configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Parameter	Description
<POLICY-NAME>	Specifies the policy to be reset.

Examples

Resetting a policy:

```
switch(config)# policy MY_POLICY reset
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show class

`show class [ip | ipv6 | mac] [<CLASS-NAME>] [commands] [configuration]`

Description

Shows class configuration information.

All parameters are optional.

Parameter	Description
`[ip	ipv6	mac]`	Selects the class type for the display: **ip** for IPv4, **ipv6** for IPv6, or **mac** for MAC classes.
`<CLASS-NAME>`	Specifies the class name.
`commands`	Specifies whether to display output as the CLI commands showing the configured class entries.
`configuration`	Specifies whether to display classes that have been configured by the user, even if they are not active due to issues with the command parameters or hardware issues. This parameter is useful during a mismatch between the entered configuration and the previous successfully programmed (active) classes.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing all class configuration:

```
switch# show class
Type Name
    Sequence Comment
           action                     L3 Protocol
           Source IP address          Source L4 Port(s)
           Destination IP address     Destination L4 Port(s)
           Additional Parameters
---------------------------------------------------------------
ipv4 MY_IPV4_CLASS
        10 my first class entry comment
           match                      icmp
           192.168.0.1/255.255.255.0
           192.168.1.1/255.255.255.0
           VLAN: 1
        20 my second class entry comment
           ignore                     tcp
           10.100.0.10/255.255.255.0  < 3000
           10.100.1.10/255.255.255.0  > 2000
           VLAN: 1
----------------------------------------------------------------------
```

Showing class configuration for the IPv4 class MY_IPV4_CLASS as CLI commands:

```
switch# show class ip MY_IPV4_CLASS commands
class ip "MY_IPV4_CLASS"
  10 match icmp 192.168.0.1/255.255.255.0 192.168.1.1/255.255.255.0 vlan 1
  10 comment my first class entry comment
  20 ignore tcp 10.100.0.10/255.255.255.0 lt 3000 10.100.1.10/255.255.255.0 gt
```

```
    2000 vlan 1
20 comment my second class entry comment
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show policy

Syntax that shows information for all policies:

```
show policy [commands] [configuration] [vsx-peer]
```

Syntax that filters by policies applied to an interface or VLAN:

```
show policy [interface <IF-NAME> [in | out | routed-in] | vlan <VLAN-ID> [in] | vni <VNI-
ID> [routed-in]]
            [commands] [configuration] [vsx-peer]
```

Syntax that filters by the named policy:

```
show policy <POLICY-NAME> [commands] [configuration] [vsx-peer]
```

Syntax that filters by the globally applied policy:

```
show policy global [commands] [configuration] [vsx-peer]
```

Syntax that shows statistical information in the form of hit counts:

```
show policy hitcounts <POLICY-NAME> [interface <IF-NAME> [in | routed-in] |
                      vlan <VLAN-ID> [in]| vni <VNI-ID> [routed-in]] [vsx-peer]
```

Syntax that shows statistical information in the form of hit counts for the globally applied policy:

```
show policy hitcounts global [vsx-peer]
```

Description

Shows information about your defined policies and where they have been applied. When **show policy** is entered without parameters, information for all policies is shown. The parameters filter the list of policies for which information is shown.

Available filtering includes:

- The content of a specific policy.
- All policies applied to a specific interface.
- All policies applied to a specific VLAN.
- All policies applied to a specific VNI.
- The globally applied policy.

To display policy statistics, use the **show policy hitcounts** form of this command.

When a policy is applied to a physical interface or lag using command **apply policy**, with the **per-interface** parameter included, unique instances of the policy are applied to each physical interface port or LAG. The unique instance of a policy has a parent-child relationship with the policy from which it was created. The **show policy** command shows information about the parent policy not the unique instances.

If a policy contains any class entries with the count keyword and policy entries with the **cir** action, and the policy is applied to multiple physical or virtual interfaces in the same direction, except for the routed ingress direction, the statistics will be aggregated. In the routed ingress direction, the statistics will be aggregated in multiple physical or virtual interfaces in the same VRF. If separate statistics for different physical or virtual interfaces are required, then another policy should be created. Alternatively, in the case of physical interfaces or LAGs, a policy applied with **per-interface** set can be used.

Parameter	Description
`interface <IF-NAME>`	Specifies the interface name.
`vlan <VLAN-ID>`	Specifies the VLAN.
`vni<VNI-ID>`	Specifies the VNI.
`in`	Selects the inbound (ingress) traffic direction.
`routed-in`	Selects the routed in traffic direction. Not applicable to a policy applied to a VLAN.
`<POLICY-NAME>`	Specifies the policy name.
`commands`	Causes the policy definition to be shown as the commands and parameters used to create it rather than in tabular form.
`configuration`	Causes the user-configured policies be shown as entered, even if the policies are not active due to policy-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) policies configuration.
`global`	Selects the globally applied policy.
`hitcounts`	Selects the policy hit counts (statistics).
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing information for all policies:

```
switch# show policy
           Name
  Sequence Comment
           Class Type
                    action
```

```
--------------------------------------------------------------------------------
           my_policy
        10 QOS class
           class1 ipv4
                    dscp af21
                    drop
        20 PBR policy.
           class2 ipv4
                    pbr mypbr
--------------------------------------------------------------------------------
```

Showing a policy as commands:

```
switch# show policy commands
policy my_policy
       10 class ip class1 action dscp af21 action drop
       20 class ip class2 action pbr mypbr
```

Showing the globally applied policy:

```
switch# show policy global commands
policy global1
    10 class ip my_class1 action drop
apply policy my_policy in
```

Showing policy hit counts (statistics) for the globally applied policy: :

```
switch# show policy hitcounts global
Statistics for Policy My_Policy:
global (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
```

Showing policy hit counts (statistics) for a policy applied everywhere (with 1/1/1 and 1/1/5 being applied per interface):

```
switch# show policy hitcounts My_Policy
Statistics for Policy My_Policy:

Interface 1/1/1,lag1 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   -  10 match tcp any any ack
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
```

```
Interface 1/1/4 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]

Interface 1/1/5 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]

interface 1/1/2.10,1/1/3.10 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   -  10 match tcp any any ack
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
...
Statistics for Policy My_Policy:

Interface 1/1/1,lag1 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   -  10 match tcp any any ack
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]


Interface 1/1/4 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]

Interface 1/1/5 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
```

```
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]

interface 1/1/2.10,1/1/3.10 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   -  10 match tcp any any ack
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
...
```

Showing policy hit counts (statistics) for a policy applied on physical interfaces and LAGs:

```
switch# show policy hitcounts My_Policy interface 1/1/1
Statistics for Policy My_Policy:

Interface 1/1/1,lag1 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
```

Showing policy hit counts (statistics) for a policy applied on VLANs:

```
switch# show policy hitcounts My_Policy vlan 30
Statistics for Policy My_Policy:

vlan 30 (in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
```

Showing policy hit counts (statistics) for a policy applied on interface VLANs:

```
switch# show policy hitcounts My_Policy interface vlan10
Statistics for Policy My_Policy:

VRF red
interface vlan 10 (routed-in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
```

```
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
```

Showing policy hit counts (statistics) for a policy applied on interface VLANs for a specific VRF:

```
switch# show policy hitcounts My_Policy vrf red routed-in
Statistics for Policy My_Policy:

VRF red
interface vlan 10 (routed-in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
Statistics for Policy My_Policy:

VRF red
interface vlan 10 (routed-in):
     Matched Packets  Configuration
10 class ip My_ip_Class
                   0  10 match tcp any any ack count
                   -  20 match udp any lt 8 any
                   0  30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
                   0  10 match tcp any any count [ 0 kbps conform ]
                   0  20 match icmpv6 1000::10 any count [ 0 kbps conform ]
```

Command History

Release	Modification
10.08	Added **[per-interface]** information. Updated examples.
10.07 or earlier	--

Command Information

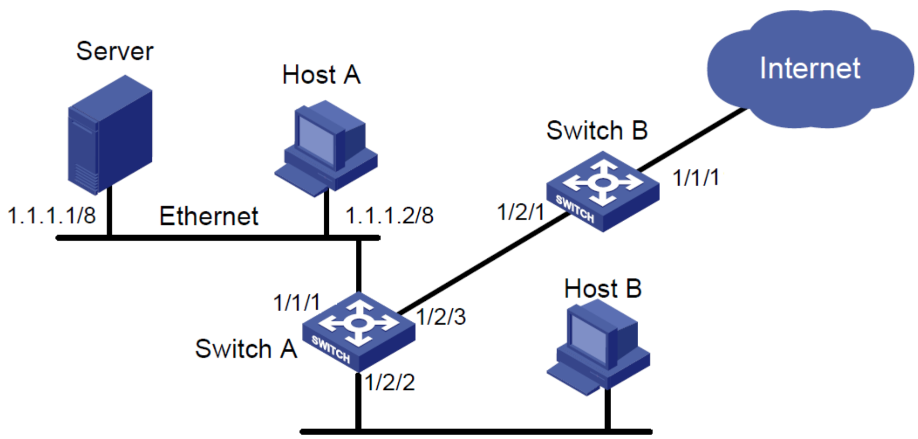
Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

Chapter 5

Classifier policies configuration example

This example configures traffic policing on:

- A 10-Gbit Ethernet of Switch A meeting the following requirements:
  - Police the rate of packets from the server to 102,400 kbps. Traffic 102,400 kbps or less is forwarded. The traffic more than 102,400 kbps is dropped.
  - Police the rate of packets from Host A to 25,600 kbps. Traffic 25,600 kbps or less is forwarded. The traffic more than 25,600 kbps is dropped.
- A 10-Gbit Ethernet 1/2/1 of Switch B limiting the incoming traffic rate of HTTP packets on 10-Gbit Ethernet 1/1/1 to the data rate of 204,800 kbps and dropping excess packets.



Configuring the classifier policies example

These steps are part of the classifier policies configuration example.

Procedure

1. Configure Switch A.

   Create traffic classes named SERVER_TRAFFIC and HOST_A_TRAFFIC for matching the packets from the server and Host A:

```
switch# configure
switch(config)# class ip SERVER_TRAFFIC
switch(config-class-ip)# match any 1.1.1.1 any
switch(config-class-ip)# exit
switch(config)# class ip HOST_A_TRAFFIC
switch(config-class-ip)# match any 1.1.1.2 any
switch(config-class-ip)# exit
```

2. Create a classifier policy named RATE_LIMIT_POLICY:

```
switch(config)# policy RATE_LIMIT_POLICY
```

3. Configure the policy RATE_LIMIT_POLICY, so that 102,400 kbps of traffic, matching the class SERVER_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy)# class ip SERVER_TRAFFIC action cir kbps 102400 exceed
drop
```

4. Configure the policy RATE_LIMIT_POLICY so that 25,600 kbps of traffic, matching the class HOST_A_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy)# class ip HOST_A_TRAFFIC action cir kbps 25600 exceed
drop
switch(config-policy)# exit
```

5. Apply RATE_LIMIT_POLICY to interface 1/1/1 for the inbound traffic:

```
switch(config)# int 1/1/1
switch(config-if)# apply policy RATE_LIMIT_POLICY in
switch(config-if)# exit
```

6. To view the configuration with the RATE_LIMIT_POLICY applied:

```
switch# show running-config
Current configuration:
!
...
class ip SERVER_TRAFFIC
    10 match any 1.1.1.1 any
class ip HOST_A_TRAFFIC
    10 match any 1.1.1.2 any
policy RATE_LIMIT_POLICY
    10 class ip SERVER_TRAFFIC action cir kbps 102400 exceed drop
    20 class ip HOST_A_TRAFFIC action cir kbps 25600 exceed drop
interface 1/1/1
    apply policy RATE_LIMIT_POLICY in
```

7. Configure Switch B.

   Create a traffic class named HTTP_TRAFFIC and configure it to match traffic to port 80:

```
switch(config)# class ip HTTP_TRAFFIC
switch(config-class-ip)# match tcp any any eq 80
switch(config-class-ip)# exit
```

8. Create a classifier policy named RATE_LIMIT_HTTP:

```
switch(config)# policy RATE_LIMIT_HTTP
```

9. Configure the policy RATE_LIMIT_HTTP so that 204,800 kbps of traffic, matching the class HTTP_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy)# class ip HTTP_TRAFFIC action cir kbps 204800 exceed
drop
switch(config-policy)# exit
```

10. Apply RATE_LIMIT_HTTP to interface 1/1/1 for inbound traffic:

```
switch(config)# int 1/1/1
switch(config-if)# apply policy RATE_LIMIT_HTTP in
switch(config-if)# exit
```

11. Show the running configuration with RATE_LIMIT_HTTP applied:

```
switch# show running-config
Current configuration:
!
...
class ip HTTP_TRAFFIC
    10 match tcp any any eq 80
policy RATE_LIMIT_HTTP
    10 class ip HTTP_TRAFFIC action cir kbps 204800 exceed drop
interface 1/1/1
    apply policy RATE_LIMIT_HTTP in
```


```
switch# show running-config
Current configuration:
!
...
class ip HTTP_TRAFFIC
    10 match tcp any any eq 80
policy RATE_LIMIT_HTTP
    10 class ip HTTP_TRAFFIC action cir kbps 204800 exceed drop
interface 1/1/1
    apply policy RATE_LIMIT_HTTP in
```

Chapter 6

ACL and Policy hardware resource

Switches have finite (TCAM and other) hardware resources used in the application of ACLs and Classifier policies to packets being processed in switch hardware. ADC (analytics data collection) also consumes TCAM lookups. Take the considerations described in this chapter into account when deciding what ACL and classifier policy-related features to use at the same time.

Show Resources

The **show resources** command allows users to see hardware resource consumption in the switch.

These hardware resources include the following:

- TCAM entries
- Policers
- Lookups
- High-Capacity TCAM entries

On the 6400 and 8400X switch series, these resources are only reported per line module.

On the 8400 switch series, the external high capacity TCAM is a dedicated hardware that can be used to improve scale. Certain IPv4 ACL types can be configured to utilize either the internal TCAM or the external high capacity TCAM. The output of **show resources** displays the current features that consume resources in the high capacity TCAM.

The usage of high capacity TCAM can be viewed specifically using **show system high-capacity-tcam**. This command displays the feature that is currently configured, pending configuration, and the default feature present in high capacity TCAM.

Event Logs

The following event logs are displayed when the TCAM runs out of resources:

Daemon	Event ID	Severity	Message
ops-switchd	10214	ERROR	TCAM Context Group selectors have been exhausted
ops-switchd	10215	ERROR	TCAM Context Group IDs have been exhausted
ops-switchd	10216	ERROR	Policer resources have been exhausted
ops-switchd	10217	ERROR	TCAM entries have been exhausted
ops-switchd	10218	ERROR	TCAM tables have

Daemon	Event ID	Severity	Message
			been exhausted
ops-switchd	10219	ERROR	TCAM ranges have been exhausted
ops-switchd	10220	ERROR	TCAM counters have been exhausted

Limitations and exclusions

On the 4100i, 6000, 6100, 8320, 8325, and 8400X switch series, resources utilized by the CoPP Lookup are created by default when the system is booted. These entries show up in the used column and may not directly correspond to configured entries.

Resources reserved for one lookup cannot be shared or used by other lookups.

TCAM resource consumption and lookups

TCAM entries and lookups are consumed by features that deal with the classification of packets using the TCAM. TCAM lookups are a finite hardware resource used in the application of ACLs and policiesto packets processed in switch hardware. ADC (analytics data collection) also consumes TCAM lookups. A switch can support a limited number of ACL and policy features at the same time. Use the command **show resources** to monitor TCAM resources and lookups.

Certain platforms have additional resources that features may consume, such as L4 port range checkers, context group selectors, policers, and counters. Different platforms consume these resources at different rates. There are a limited number of features that use TCAM that can be enabled simultaneously on the same line card.

In the following TCAM lookup lists, **IP** means both IPv4 and IPv6.

TCAM lookup behaviors vary by switch type.

For the 8400 Switch series:

The Ingress TCAM has eight lookups split into two cycles. Each individual lookup can be configured to contain either single or double width TCAM entries. Only one lookup is used to contain double width TCAM entries. However, it takes *two* lookups to contain *quadruple* width TCAM entries, and both of the two lookups must be in the same cycle.

Entry width	Number of required lookups
1	1
2	1
4	2 (must be in the same cycle)

A quadruple entry width ingress TCAM group can fail to install if they are in different cycles. AOS-CX avoids this issue at boot using the following process:

1. AOS-CX attempts to install quadruple width ingress TCAM groups first in the first cycle, then the second if the first is full.
2. AOS-CX attempts to install single- and double-width ingress TCAM groups first in the second cycle, then the first if the second is full.

The ingress TCAM groups in a startup-config that were previously successful should correctly install as a result of this process. However, a user may run into this issue while making config changes during runtime.

The ingress TCAM preselectors allow the ingress TCAM lookups to have different configurations for mutually exclusive traffic. This allows IPv4 and IPv6 ingress TCAM groups to share these hardware resources. The following table lists the Ingress TCAM preselector for each feature.

Stage	Feature name	Width	Lookups	Preselector
Ingress	Ingress MAC Port ACL	4	2	Physical interfaces
Ingress	Ingress MAC VLAN ACL	4	2	Physical interfaces
Ingress	Ingress IPv4 Port ACL	4	2	IPv4 packets on physical interfaces
Ingress	Ingress IPv4 VLAN ACL	4	2	IPv4 packets on Physical Interfaces
Ingress	Routed Ingress IPv4 VLAN ACL	4	2	Routed IPv4 packets on physical interfaces
Ingress	Ingress IPv6 Port ACL	4	2	IPv6 packets on physical interfaces
Ingress	Ingress IPv6 VLAN ACL	4	2	IPv6 packets on physical interfaces
Ingress	Routed Ingress IPv6 VLAN ACL	4	2	Routed IPv6 packets on physical interfaces
Ingress	Egress ACL Logging*	2	1	Recycle port
Ingress	Ingress IPV4 High Capacity ACL Result	1	1	IPv4 packets on physical interfaces
Ingress	Ingress MAC+IPv4 Port Policy	4	2	Physical interfaces
Ingress	Ingress MAC+IPv4 VLAN Policy	4	2	Physical interfaces
Ingress	Ingress MAC+IPv4 Global Policy	4	2	Physical interfaces
Ingress	Ingress IPv6 Global Policy	4	2	IPv6 packets on physical interfaces
Ingress	Routed Ingress IPv4 Port Policy	4	2	Routed IPv4 packets on physical interfaces
Ingress	Routed Ingress IPv4 VLAN Policy	4	2	Routed IPv4 packets on physical interfaces

Stage	Feature name	Width	Lookups	Preselector
Ingress	Ingress IPv6 Port Policy	4	2	IPv6 packets on physical interfaces
Ingress	Ingress IPv6 VLAN Policy	4	2	IPv6 packets on physical interfaces
Ingress	Routed Ingress IPv6 Port Policy	4	2	Routed IPv6 packets on physical interfaces
Ingress	Routed Ingress IPv6 VLAN Policy	4	2	Routed IPv6 packets on physical interfaces
Ingress	MAC Control Plane Policing	2	1	Physical interfaces
Ingress	IPV4 Control Plane Policing	2	1	IPv4 packets on physical interfaces
Ingress	IPV6 Control Plane Policing	2	1	IPv6 packets on physical interfaces
Ingress	IPv6 Tunnel Control Plane Policing	2	1	IPv4 or IPv6 packets on physical interfaces
Ingress	VXLAN Tunnel Control Plane Policing	4	2	Physical interfaces
Ingress	Ingress MAC VSX	1	1	All traffic
Ingress	Ingress IPv4 VSX	2	1	IPv4 packets
Ingress	Ingress IPv6 VSX	2	1	IPv6 packets
Ingress	Routed Ingress Unicast IPv4 Counter	1	1	IPv4 packets on physical interfaces
Ingress	Routed Ingress Unicast IPv6 Counter	1	1	IPv6 packets on physical interfaces
Ingress	Bidrectional Forwarding Detection (BFD)	2	1	IPv4 packets on physical interfaces
Ingress	Ingress IPv64 Analytics Data Collection	2	1	IPv4 packets on physical interfaces
Ingress	Ingress IPv6 Analytics Data Collection	4	2	IPv6 packets on physical interfaces
Ingress	Ingress IPv4 IPLD	4	2	IPv4 packets on physical interfaces
Ingress	Ingress IPv6 IPLD	4	2	IPv6 packets on physical interfaces
Ingress	Ingress IPv4 ECN	1	1	IPv4 packets on physical interfaces
Ingress	Ingress IPv6 ECN	1	1	IPv6 packets on physical

Stage	Feature name	Width	Lookups	Preselector
				interfaces
Ingress	L2 VXLAN DSCP COPY	1	1	Bridged IPv4 packets on physical interfaces

The following ingress TCAM groups are installed by default:

- Ingress IPV4 High Capacity ACL Result (needed by the High Capacity TCAM)
- MAC Control Plane Policing
- IPV4 Control Plane Policing
- IPV6 Control Plane Policing

Examples

For the following config:

```
access-list ip test
class ip test
     10 match any any any
policy p
     10 class ip test action drop
vlan 10
     apply access-list ip test in
interface vlan 10
     apply policy p routed-in
interface 1/1/1
     no shutdown
     l3-counters rx
```

Here is how the lookups will be allocated:

Table 1: *Cycle 1*

Preselector	Lookup 1	Lookup 2	Lookup 3	Lookup 4
Routed & IPv6 & card ports				
Bridged & IPv6 & card ports				
Routed & IPv4 & card ports	Ingress IPv4 VLAN ACL	Ingress IPv4 VLAN ACL	Routed Ingress IPv4 VLAN policy	Routed ingress IPv4 VLAN policy
Bridged & IPv4 & card ports	Ingress IPv4 VLAN ACL	Ingress IPv4 VLAN ACL		
Card ports				
Recycle ports				
Default				

Table 2: *Cycle 2*

Preselector	Lookup 1	Lookup 2	Lookup 3	Lookup 4
Routed & IPv6 & card ports		MAC Control Plane Policing	IPv6 Control Plane Policing	Routed Ingress Unicast IPv6 Counter
Bridged & IPv6 & card ports		MAC Control Plane Policing	IPv6 Control Plane Policing	Routed Ingress Unicast IPv6 Counter
Routed & IPv4 & card ports	Ingress IPv4 High Capacity ACL result	MAC Control Plane Policing	IPv4 Control Plane Policing	Routed ingress IPv4 VLAN policy
Bridged & IPv4 & card ports	Ingress IPv4 VLAN ACL	MAC Control Plane Policing	IPv4 Control Plane Policing	Routed ingress IPv4 VLAN policy
Card ports		MAC Control Plane Policing		
Recycle ports				
Default		MAC Control Plane Policing		

The **routed & IPv4 & card ports** preselector is full so no more ingress TCAM groups that match on that set of traffic can be installed. However, there are five available lookups for ingress TCAM groups that only match on **IPv6 & card** port traffic. There is also room for the ingress TCAM group that matches on the recycle port which is used to make ACL Logging work.

Here's an example config that takes advantage of most of the **IPv6 & card ports** lookups and shows that the **routed & IPv4 & card ports** preselector is full:

```
access-list ip test
      10 deny any any any log count
access-list ipv6 testv6
class ip test
      10 match any any any
class ipv6 test
      10 match any any any
policy p
      10 class ip test action drop
      20 class ipv6 test action drop
policy p_v4_only
      10 class ip test action drop
vlan 10
     apply access-list ip test in
     apply access-list ipv6 testv6 in
interface vlan 10
     apply policy p routed-in
interface 1/1/1
     no shutdown
     l3-counters rx
     apply access-list ip test out
! policy p_v4_only user configuration does not match active configuration.
! run 'policy NAME reset' to reset policy to match active configuration.
apply policy p_v4_only routed-in
```

Here is how the lookups will be allocated:

Table 3: *Cycle 1*

Preselector	Lookup 1	Lookup 2	Lookup 3	Lookup 4
Routed & IPv6 & card ports	Ingress IPv6 VLAN ACL	Ingress IPv6 VLAN ACL	Routed Ingress IPv6 VLAN Policy	Routed Ingress IPv6 VLAN Policy
Bridged & IPv6 & card ports	Ingress IPv6 VLAN ACL	Ingress IPv6 VLAN ACL		
Routed & IPv4 & card ports	Ingress IPv4 VLAN ACL	Ingress IPv4 VLAN ACL	Routed Ingress IPv4 VLAN Policy	Routed ingress IPv4 VLAN policy
Bridged & IPv4 & card ports	Ingress IPv4 VLAN ACL	Ingress IPv4 VLAN ACL		
Card ports				
Recycle ports				
Default				

Table 4: *Cycle 2*

Preselector	Lookup 1	Lookup 2	Lookup 3	Lookup 4
Routed & IPv6 & card ports		MAC Control Plane Policing	IPv6 Control Plane Policing	Routed Ingress Unicast IPv6 Counter
Bridged & IPv6 & card ports		MAC Control Plane Policing	IPv6 Control Plane Policing	Routed Ingress Unicast IPv6 Counter
Routed & IPv4 & card ports	Ingress IPv4 High Capacity ACL Result	MAC Control Plane Policing	IPv4 Control Plane Policing	Routed Ingress Unicast IPv4 Counter
Bridged & IPv4 & card ports	Ingress IPv4 High Capacity ACL Result	MAC Control Plane Policing	IPv6 Control Plane Policing	Routed Ingress Unicast IPv4 Counter
Card ports		MAC Control Plane Policing		
Recycle ports	Egress ACL Logging			
Default		MAC Control Plane Policing		

Note: Port IPv4 ACLs and features applied on egress use dedicated hardware and do not conflict with the lookup usage of any of the previous features. However, the TCAM entries are in a shared space between ingress and egress, *except* for the High-capacity TCAM entries.

Egress features such as IPv4 ACLs applied "out" to route-only ports use dedicated lookup hardware, but they share TCAM entries with ingress features.

TCAM banks

There are 12 large banks of 4k single width TCAM entries and 4 small TCAM banks of 256 single width TCAM entries. Double width TCAM entries cut the TCAM bank sizes in half. Quadruple width TCAM entries can span two consecutive banks (even and odd). For example, bank-ID 0 can only be combined with bank-ID 1 to be spanned by quadruple width TCAM entries. A TCAM bank can be assigned to either the Ingress TCAM or Egress TCAM.

The following table shows TCAM bank entries. For entries denoted with an *, the odd numbered bank was combined with the even numbered bank preceeding it for quadruple width TCAM entries to span.

Bank-ID	Single width entries	Double width entries	Quadruple width entries
0	4,096	2,048	2,048
1	4,096	2,048	*
2	4,096	2,048	2,048
3	4,096	2,048	*
4	4,096	2,048	2,048
5	4,096	2,048	*
6	4,096	2,048	2,048
7	4,096	2,048	*
8	4,096	2,048	2,048
9	4,096	2,048	*
10	4,096	2,048	2,048
11	4,096	2,048	*
12	256	128	128
13	256	128	*
14	256	128	128
15	256	128	*
Total	50,176	25,088	12,544

TCAM bank sharing

A TCAM bank can house TCAM entries from more than one TCAM group if all of the following are true:

- The TCAM groups must be mutually exclusive (for example, IPv4 and IPv6).
- The Ingress TCAM groups must be in the same cycle of the Ingress TCAM.
- The entry width must be same (single/double/quadruple).
- The keys must have enough space to add a unique ID (up to four bits) which are added by the preselectors to identify which TCAM group the packet is for.

TCAM bank sharing can cause TCAM group install order issues. This occurs when two or more TCAM groups share a bank in one install order but do not share in a different install order. This can happen when TCAM groups that can share banks are installed in different cycles in the Ingress TCAM or when different TCAM groups share a bank with different TCAM groups due to install order.

The show resources output may be incorrect for TCAM bank sharing. The output assumes specific TCAM groups share with each other based on precedent, but that may differ from what is in hardware.

TCAM bank allocation

There are two different TCAM bank allocation methods for TCAM groups on the 8400 because there are small and large TCAM banks. The allocation methods are as follows:

- Large bank preference: These TCAM groups first try to allocate large banks. Once the large banks are all allocated, small banks are allocated.
- Small bank preference: These TCAM groups first try to allocate small banks. As of CX 10.10.1090, once the small banks are allocated, large banks are allocated.

All TCAM groups have a large bank preference except for the following, which have small bank preference:

- MAC Control Plane Policing
- IPv4 Control Plane Policing
- IPv6 Control Plane Policing
- IPv6 Tunnel Control Plane Policing
- VXLAN Tunnel Control Plane Policing
- Ingress MAC VSX
- Ingress IPv4 VSX
- Ingress IPv6 VSX
- Egress VSX

Matching precedence order

When a packet is matched by multiple TCAM Lookups with the same action, a precedence order is followed.

For example, if a packet matches an IPv6 ACL with a count action and a MAC ACL with a count action, the IPv6 count action takes precedence and the MAC ACL will not count the packet. However, if a packet matches both an ACL and a policy with count actions, both will be counted. Regardless of precedence, if a packet is to be dropped by a configured feature, it will be dropped. Ingress packets do not take precedence over egress packets nor do egress packets take precedence over ingress packets.

Applies only to deny**, not** permit: If a policy counts and mirrors a packet which also matches an ACL that counts the packet, the hitcounts for the ACL and policy will not increment. Mirrored packets should configure only one counter, in either the ACL or the policy, but not both.

The precedence order from highest to lowest is as follows:

```
Ingress Port IPv6 ACL
Ingress VLAN IPv6 ACL
Ingress Routed VLAN IPv6 ACL
Ingress Port IPv4 ACL
Ingress Routed VLAN IPv4 ACL
Ingress VLAN IPv4 ACL
Ingress Port MAC ACL
Ingress VLAN MAC ACL
Ingress Port Policy with IPv6 classes
Ingress Port Policy with IPv4 and/or MAC classes
Ingress VLAN Policy with IPv6 classes
Ingress VLAN Policy with IPv4 and/or MAC classes
Bidirectional Forwarding Detection (BFD)
Ingress IPv4 VSX
Ingress IPv6 VSX
Ingress MAC VSX
IPv6 Control Plane Policing
IPv4 Control Plane Policing
MAC Control Plane Policing
Ingress Routed IPv6 Port Policy
Ingress Routed IPv4 Port Policy
Ingress Routed IPv6 VLAN Policy
Ingress Routed IPv4 VLAN Policy
Ingress Routed IPv6 Unicast Counters
Ingress Routed IPv6 Multicast Counters
Ingress Routed IPv4 Unicast Counters
Ingress Routed IPv4 Multicast Counters
Ingress IPv6 Analytics Data Collection (ADC)
Ingress IPv4 Analytics Data Collection (ADC)
Egress VSX
Egress Routed IPv4 Port ACL
Egress Routed IPv6 Unicast Counters
Egress Routed IPv6 Multicast Counters
Egress Routed IPv4 Unicast Counters
Egress Routed IPv4 Multicast Counters
```

L4 port ranges

On platforms that support the use of dedicated resources called **L4 Port Ranges**, any ACE that uses **lt**, **gt**, **range**, or port groups attempts to use these dedicated hardware resources. L4 port ranges are used to reduce the number of TCAM entries needed to cover an L4 port range. For example, range `50000 50005` could be covered by a single L4 port range or by two TCAM entries using the following value/masks:

Entry number	L4 port range	Value/Mask
1	range 50000 50003	0xC350/0xFFFC
2	range 50004 50005	0xC354/0xFFFE

The problem gets multiplied when an ACE matches on two L4 port ranges. The TCAM entries used to cover a source and destination L4 port range must cover all possibilities. For example, for this ACE,

```
10 permit tcp any range 100 200 any range 50000 50005
```

The `range 100 200` needs six value masks. As a result, this ACE needs 12 (6*2=12) TCAM entries. However, if the TCAM group uses L4 port ranges then the ACE only needs one TCAM entry that matches on two L4 port ranges.

On the 8400,switch series, TCAM entries can share an L4 port range given the following conditions:

- The min and max of the L4 port range are the same
- The L4 port range type (either source or destination L4 ports) is the same
- The TCAM entries are on the same ingress packet processing pipeline

The TCAM entries do not need to be in the same TCAM group.

L4 port ranges are not allocated if the L4 port range can be covered by a single value/mask. For example, **eq 1** and **lt 1024** do not allocate an L4 port range. As a result, the L4 port range allocation can be reduced if the L4 port range is split into multiple ACEs that can each be covered by a single value/mask. For example, the following ACE:

```
10 permit tcp any any range 50000 50005
```

can be split into these two ACEs with L4 port ranges that can each be covered by a single value/mask:

```
10 permit tcp any any range 50000 50003
11 permit tcp any any range 50004 50005
```

Reducing the number of L4 port ranges needed by a config can be desirable if the config requires more L4 port ranges than are available in the hardware and there are available TCAM entries to absorb the additional ACEs.

On the 8400 switch series, there are 24 L4 port ranges per ingress TCAM. L4 port ranges are used to match on consecutive ranges of L4 ports.

The following TCAM groups use L4 port ranges:

Stage	Feature name
Ingress	Ingress IPv4 Port ACL (when not in high-capacity TCAM)
Ingress	Ingress IPv4 VLAN ACL (when not in high-capacity TCAM)
Ingress	Routed Ingress IPv4 VLAN ACL (when not in high-capacity TCAM)
Ingress	Ingress MAC+IPv4 Port Policy
Ingress	Ingress MAC+IPv4 VLAN Policy
Ingress	Ingress MAC+IPv4 Global Policy
Ingress	Routed Ingress IPv4 Port Policy
Ingress	Routed Ingress IPv4 VLAN Policy

Any unsupported feature may use more than one hardware entry to represent the range of L4 ports.

ACL and Policy hardware resource commands

show resources

```
show resources [<SLOT-ID>] [vsx-peer]
```

Description

Shows hardware resource consumption for the specified line modules or for all line modules. Resource data is updated every 10 seconds.

Hardware resource consumption information is shown for:

- TCAM entries
- Policers
- L4 Port Ranges

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the line module. For example, to specify the line module in member 1, slot 2, enter `1/2`.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing hardware resource consumption on a 8400X switch:

```
switch# show resources

Resource Usage:

Mod  Description
         Resource                                Width    Used   Reserved   Free
-------------------------------------------------------------------------------
1/1  Ingress IPv4 Port ACL
         High-Capacity TCAM/LPM Entries          2        0      262144
     MAC Control Plane Policing
         TCAM Entries                            2        16     256
     IPv4 Control Plane Policing
         TCAM Entries                            2        70     256
     IPv6 Control Plane Policing
         TCAM Entries                            2        72     *
     IPv4 Unicast Route
         High-Capacity TCAM/LPM Entries          1        0      131072
     IPv6 Unicast Route
         High-Capacity TCAM/LPM Entries          2        0      262144
     IPv4 Multicast Route
         High-Capacity TCAM/LPM Entries          2        0      65536
     IPv6 Multicast Route
         High-Capacity TCAM/LPM Entries          4        0      65536
     Total
         TCAM Entries                                     158    512        49664
         High-Capacity TCAM/LPM Entries                   0      786432     258048
         Policers                                         0                 65536
         Ingress L4 Port Ranges                           0                 24
*This feauture shares reserved resources with the preceeding feature.
```

Showing hardware resource consumption for line module 1/1 on a 8400X switch:

```
switch# show resources 1/1
```

```
Resource Usage:

Mod  Description
         Resource                          Width    Used Reserved    Free
-------------------------------------------------------------------------
1/1  Ingress IPv4 Port ACL
         High-Capacity TCAM/LPM Entries        2       0   262144
     MAC Control Plane Policing
         TCAM Entries                          2      16      256
     IPv4 Control Plane Policing
         TCAM Entries                          2      70      256
     IPv6 Control Plane Policing
         TCAM Entries                          2      72        *
     IPv4 Unicast Route
         High-Capacity TCAM/LPM Entries        1       0   131072
     IPv6 Unicast Route
         High-Capacity TCAM/LPM Entries        2       0   262144
     IPv4 Multicast Route
         High-Capacity TCAM/LPM Entries        2       0    65536
     IPv6 Multicast Route
         High-Capacity TCAM/LPM Entries        4       0    65536
     Total
         TCAM Entries                                158      512   49664
         High-Capacity TCAM/LPM Entries                0   786432  258048
         Policers                                      0            65536
         Ingress L4 Port Ranges                        0               24
* This feature shares reserved resources with the preceding feature.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	

Chapter 7

Configurable high capacity TCAM

The high capacity TCAM supports up to 512k entries. The standard TCAM supports up to 12k entries.

Previously, the higTCAM Resource Utilization Monitor NAE Agenth capacity TCAM was dedicated to **Ingress IPv4 Port ACLs**. It is now possible to reconfigure the high capacity TCAM to instead be used for one of two other IPv4 ingress ACL types.

- **Ingress IPv4 Port ACLs** (the default)
- **Ingress IPv4 VLAN ACLs**
- **Routed Ingress IPv4 VLAN ACLs**

This makes it possible for you to configure your TCAM capacity according to the type of IPv4 ACLs that you expect to use most heavily. It is expected that this type of configuration would be performed infrequently.

Upon reboot, the high capacity TCAM is entirely reconfigured for the selected IPv4 ingress ACL type, causing the previous ACL type to now only use the standard TCAM.

If any of the ACLS of the type moved to the standard TCAM are too large to have their ACEs programmed in their entirety due to lack of space in the standard TCAM, such ACLs will be left unapplied.

Configuring the high capacity TCAM

Configuring the high capacity TCAM requires a reboot of either the switch chassis or all line modules so this configuration change should be scheduled for a maintenance window.

Performing the configuration

Issue the **system high-capacity-tcam access-list ip** command, selecting one of three IPv4 ingress ACL types:

- **port in** for Ingress IPv4 Port ACLs (the default)
- **vlan in** for Ingress IPv4 VLAN ACLs
- **vlan routed-in** for Routed Ingress IPv4 VLAN ACLs

For example, configuring Ingress IPv4 VLAN ACLs:

```
switch(config)# system high-capacity-tcam access-list ip vlan in
This command changes how the high capacity TCAM is used and requires
a reboot to complete. After reboot, any entries that were using the
high capacity TCAM will be moved to lower capacity hardware and may
no longer fit.

Continue (y/n)? y
switch(config)#
```

Showing the configuration before reboot. Notice the indicated Configured feature shows **Ingress IPv4 VLAN ACL** and the reminders to reboot.

```
switch# show system high-capacity-tcam
High-Capacity TCAM/LPM Usage:
Configured feature : Ingress IPv4 VLAN ACL
Default feature    : Ingress IPv4 Port ACL

Module    Current Feature
-------------------------------
1/1       Ingress IPv4 Port ACL  (reboot required to apply new configuration)
1/4       Ingress IPv4 Port ACL  (reboot required to apply new configuration)
```

Verifying the updated configuration

After either rebooting the entire switch or all of its line modules, verify that the updated configuration is now in effect. Notice how the selected ACL type **Ingress IPv4 VLAN ACLs** is now shown for all line modules and matches the indicated **Configured feature**:

```
switch# show system high-capacity-tcam
High-Capacity TCAM/LPM Usage:
Configured feature : Ingress IPv4 VLAN ACL
Default feature    : Ingress IPv4 Port ACL

Module    Current Feature
-------------------------------
1/1       Ingress IPv4 VLAN ACL
1/4       Ingress IPv4 VLAN ACL
```

The **show resources** command can be used to see that the Ingress IPv4 VLAN ACL is now assigned to the high capacity TCAM (bolded in this sample output):

```
switch# show resources 1/1

Resource Usage:

Mod  Description
         Resource                          Width    Used Reserved    Free
-------------------------------------------------------------------------
1/1  Ingress IPv4 Port ACL
         TCAM Entries                          4      16     8192
     Ingress IPv4 VLAN ACL
         High-Capacity TCAM/LPM Entries        2       0   262144
     MAC Control Plane Policing
         TCAM Entries                          2      18      256
     IPv4 Control Plane Policing
         TCAM Entries                          2      80      256
     IPv6 Control Plane Policing
         TCAM Entries                          2      78        *
     Total
         TCAM Entries                               176      512   41472
         High-Capacity TCAM/LPM Entries               0     4096 1044480
         Policers                                     0            65536
         Ingress L4 Port Ranges                       0               24
* This feature shares reserved resources with the preceding feature.
```

VSX sync requirements

In a VSX environment the high capacity TCAM configuration must be synchronized to the secondary switch and both switches (or all line cards in both switches) must be rebooted.

Running the `vsx-sync` command on the primary switch:

```
switch-primary(config)# vsx
switch-primary(config-vsx)# vsx-sync hardware-high-capacity-tcam
```

Showing the high capacity TCAM configuration on the secondary switch. Notice the indicated Configured feature shows `Ingress IPv4 VLAN ACL` and the reminders to reboot.

```
switch-secondary(config)# show system high-capacity-tcam
High-Capacity TCAM/LPM Usage:
Configured feature : Ingress IPv4 VLAN ACL
Default feature    : Ingress IPv4 Port ACL

Module    Current Feature
-------------------------------
1/1       Ingress IPv4 Port ACL  (reboot required to apply new configuration)
1/4       Ingress IPv4 Port ACL  (reboot required to apply new configuration)
```

After either rebooting the entire secondary switch or all of its line modules, verify that the updated configuration is now in effect. Notice how the selected ACL type `Ingress IPv4 VLAN ACLs` is now shown for all line modules and matches the indicated `Configured feature`:

```
switch-secondary# show system high-capacity-tcam
High-Capacity TCAM/LPM Usage:
Configured feature : Ingress IPv4 VLAN ACL
Default feature    : Ingress IPv4 Port ACL

Module    Current Feature
-------------------------------
1/1       Ingress IPv4 VLAN ACL
1/4       Ingress IPv4 VLAN ACL
```

High capacity TCAM commands

system high-capacity-tcam access-list ip

```
system high-capacity-tcam access-list ip port in
system high-capacity-tcam access-list ip vlan {in|routed-in}

no system high-capacity-tcam access-list ip port in
no system high-capacity-tcam access-list ip vlan {in|routed-in}
```

Description

Configures the high capacity TCAM for one of three types of IPv4 ingress ACLs. Full switch chassis reboot or reboot of every line module is required before the configuration change takes effect.

The **no** form of this command reverts the high capacity TCAM to its default of Ingress IPv4 Port ACLs.

Parameter	Description
`port in`	Selects Ingress IPv4 Port ACLs. This is the default.
`vlan in`	Selects Ingress IPv4 VLAN ACLs.
`vlan routed-in`	Selects Routed Ingress IPv4 VLAN ACLs.

Usage

- Upon reboot, the high capacity TCAM is entirely reconfigured for the selected IPv4 ingress ACL type, causing the previous ACL type to now only use the standard internal TCAM with its smaller capacity.

In a VSX environment the high capacity TCAM configuration must be synchronized to the secondary switch and both switches (or all line cards in both switches) must be rebooted before the new configuration takes effect on either switch.

Examples

Configuring the high capacity TCAM for Ingress IPv4 VLAN ACLs:

```
switch(config)# system high-capacity-tcam access-list ip vlan in
This command changes how the high capacity TCAM is used and requires
a reboot to complete. After reboot, any entries that were using the
high capacity TCAM will be moved to lower capacity hardware and may
no longer fit.

Continue (y/n)? y
switch(config)#
```

Reverting the high capacity TCAM to its default of Ingress IPv4 Port ACLs:

```
switch(config)# no system high-capacity-tcam access-list ip vlan in
This command changes how the high capacity TCAM is used and requires
a reboot to complete. After reboot, any entries that were using the
high capacity TCAM will be moved to lower capacity hardware and may
no longer fit.

Continue (y/n)? y
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution rights for this command.

show system high-capacity-tcam

`show system high-capacity-tcam [commands]`

Description

Shows the high capacity TCAM feature current configuration or the pending configuration change (if any) and the current high capacity TCAM feature configuration of each line module plus a reminder to reboot (when needed). The three possible high capacity TCAM feature names are displayed as follows (**Ingress IPv4 Port ACL** is the default):

```
Ingress IPv4 Port ACL
Ingress IPv4 VLAN ACL
Routed Ingress IPv4 VLAN ACL
```

Parameter	Description
`commands`	Changes the **show** command output to be in the form of CLI commands.

Examples

Configuring the high capacity TCAM for **Ingress IPv4 VLAN ACLs** and then showing the configuration before and after reboot. Notice the reminders for the required reboot.

```
switch(config)# system high-capacity-tcam access-list ip vlan in
This command changes how the high capacity TCAM is used and requires
a reboot to complete. After reboot, any entries that were using the
high capacity TCAM will be moved to lower capacity hardware and may
no longer fit.

Continue (y/n)? y
switch(config)#
switch(config)# show system high-capacity-tcam
High-Capacity TCAM/LPM Usage:
Configured feature : Ingress IPv4 VLAN ACL
Default feature    : Ingress IPv4 Port ACL

Module    Current Feature
-------------------------------
1/1       Ingress IPv4 Port ACL  (reboot required to apply new configuration)
1/4       Ingress IPv4 Port ACL  (reboot required to apply new configuration)


...
## after the reboot of the entire switch or all line modules ##

switch# show system high-capacity-tcam
High-Capacity TCAM/LPM Usage:
Configured feature : Ingress IPv4 VLAN ACL
Default feature    : Ingress IPv4 Port ACL

Module    Current Feature
-------------------------------
1/1       Ingress IPv4 VLAN ACL
1/4       Ingress IPv4 VLAN ACL
```

Showing the configuration change as commands:

```
switch# show system high-capacity-tcam commands
system high-capacity-tcam access-list ip vlan in
! not applied to module 1/1 - reboot required
! not applied to module 1/4 - reboot required
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manager (#)	

Chapter 8

Support and Other Resources

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html
AOS-CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

HPE Aruba Networking Hardware Documentation and Translations Portal	https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htmm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at https://networkingsupport.hpe.com.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to https://www.arubanetworks.com/support-services/product-warranties/.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see https://www.arubanetworks.com/company/about-us/environmental-citizenship/.

Documentation Feedback

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